A Little Sedge Goes a Long Way:

Emergent Vegetation as a Key Driver of Littoral Methane Flux

by

Kelsey McGuire

B.Sc., The University of British Columbia, 2024

A THESIS SUBMITTED IN PARTIAL FULFILLMENT OF

THE REQUIREMENTS FOR THE DEGREE OF

MASTER OF SCIENCE

in

THE FACULTY OF GRADUATE AND POSTDOCTORAL STUDIES

(Geography)

THE UNIVERSITY OF BRITISH COLUMBIA

(Vancouver)

August 2026

The following individuals certify that they have read, and recommend to the Faculty of Graduate and Postdoctoral Studies for acceptance, the thesis entitled:

A Little Sedge Goes a Long Way: Emergent Vegetation as Key Driver of Littoral Methane Flux

submitted by Kelsey McGuire in partial fulfilment of the requirements for

the degree of Master of Science

in Geography

Examining Committee:

Dr. McKenzie Kuhn, Assistant Professor, Department of Geography, UBC
Supervisor

Dr. Mark Johnson, Professor, IRES & EOSC Departments, UBC
Supervisory Committee Member

Dr. Marysa Laguë, Assistant Professor, Department of Geography, UBC
Supervisory Committee Member

Abstract

In northern (>50°N) freshwater ecosystems, aquatic vegetation is expanding, potentially enhancing methane (CH_4) emissions through plant-mediated transport and organic matter input. Despite this, most studies do not directly measure CH_4 flux from shoreline emergent vegetation. To understand influences of emergent water sedges (*Carex aquatilis*) on CH_4 emissions, we measured diffusive, ebullitive, and plant-mediated flux along a vegetated edge of a large, post-glacial lake in Kaska Dene ancestral territory (now called Northern British Columbia). Chamber-based CH_4 fluxes from vegetated zones (86.2 ± 57.3 mg C m^{-2} d^{-1}; mean ± std; diffusion + plant mediated) were 3.8 times greater than diffusive fluxes from adjacent open areas (22.7 ± 19.5 mg C m^{-2} d^{-1}). Ebullitive fluxes were similar across vegetated and open water areas, emitting 18.1 ± 28.7 and 22.0 + 28.0 mg C m^{-2} d^{-1}, respectively. Using temperature incubations at 20 and 30°C we found that treatments with benthic vegetation had the higher cumulative potential CH_4 production rates (52.2 and 252 nmol CH_4 per grams dry weight (g DW)) and temperature sensitivity (Q_{10} = 9.8) compared to sediment (37.6 and 35.7 nmol CH_4 g DW^{-1}; Q_{10} = 4.7) and sediment + benthic vegetation (39.2 and 109 nmol CH_4 g DW^{-1}; Q_{10} = 7.4) treatments. Our results suggest that as aquatic vegetation expands in northern lakes, we are likely to see increased CH_4 emissions facilitated by plant transport and inputs of more labile organic matter, increasing the temperature sensitivity of CH_4 production.

Lay Summary

Under global warming, we are witnessing the expansion of aquatic vegetation into northern lakes, which has repercussions for methane (CH_4), a potent greenhouse gas that is naturally produced in aquatic environments. In this thesis I sought to explore how shoreline vegetation alters the production and exchange of CH_4 within a Kaska Dene lake (Northern British Columbia). We found that vegetated shorelines have nearly four times the amount of CH_4 being released than nearby open water areas. Vegetation not only promotes greater direct release of CH_4, but using lab-based incubations, I found that submerged mosses produced higher amounts of CH_4 in comparison to vials with just lake sediment. My incubations also revealed that mosses were highly sensitive to temperature, producing more CH_4 when exposed to warmer temperatures. My results suggest that under lake warming and subsequent vegetation expansion, CH_4 emission from northern lakes are likely to increase.

Preface

I contributed to project creation, project logistics, data collection and analysis, and written thesis development and revision, under the supervision of Dr. McKenzie Kuhn. Dr. Mark Johnson provided feedback during the development of the methodology, and both he and Dr. Marysa Laguë provided final thesis revisions.

Lyreshka Castro Morales and Sumaiyah Choudry contributed to data collection in the field and laboratory. Himari Honda supported statistical analyses and project development. Isotope analyses were conducted by Dr. Clarice Perryman at Standford University's Terrestrial Carbon Cycle Group, surface water chemistry analyses by Allan Harms at the University of Alberta's Natural Resources Analytical Laboratory, and Loss on Ignition and Carbon/Nitrogen analyses were done at the University of British Columbia within the Geography and the Earth, Ocean and Atmospheric Sciences (Jian Guo) departments respectively.

I am primarily responsible for data collection, sample preparation, data analysis, and manuscript composition. Chapter 2 will be prepared for peer-review, with intention for publication. Generative AI tools, such as ChatGPT and Google Gemini, were restricted to helping in the creation and debugging of R code. Both tools were used for tasks such as giving suggestions for optimization of functions and debugging and corrections for final plot design.

Table of Contents

List of Tables

List of Figures

List of Symbols

Symbol	Name
δ	delta

List of Abbreviations

Abbreviation	Full Form	Units
ABCFLUX V2	Arctic–Boreal CO_2 and CH_4 Flux	NA
BAWLD-CH_4	Boreal-Arctic Wetland and Lake Dataset – CH_4	NA
C	Carbon	mg C m^{-2} d^{-1}
CH_4	Methane	mg C m^{-2} d^{-1}
CO_2	Carbon Dioxide	mg C m^{-2} d^{-1}
C/N	Carbon to Nitrogen Ratio	NA
DO	Dissolved Oxygen	%
DOC	Dissolved Organic Carbon	mg L^{-1}
emmeans	Estimated Marginal Means	NA
GCC	Greenness Chromatic Coordinate	NA
GLMM	Generalized Linear Mixed Model	NA
LOI	Loss on Ignition	NA
N_2	Nitrogen	NA
NH_4-N	Ammonium	µg L^{-1}
O_2	Oxygen	NA
OM	Organic Matter	% or g DW^{-1}
PO_4^{3-}	Phosphate	µg L^{-1}
TDS	Total Dissolved Solids	mg L^{-1}
TN	Total Nitrogen	µg L^{-1}
TOC	Total Organic Carbon	mg L^{-1}

Acknowledgements

Writing this section has made me realize just how much love I hold for the community I have in my life. There are so many people I would like to thank, and apologies if the speech cut-off song plays at any point during this.

First, to my supervisor Dr. McKenzie Kuhn, I am truly grateful to have the opportunity of working with and learning from you throughout my degree. I don't know if this experience would have been as wonderful if you weren't a wonderful person yourself, and your kindness and support will never be forgotten, especially concerning unit conversions. To the BABs, I want to thank you for being wonderful lab members and always being willing to entertain my non-science conversations – team spoon the whole way.

To my friends and family, who are an incredible support system, I thank you from the bottom of my heart. To Himari, thank you so much for your friendship, I don't even want to think about what life would be like if we hadn't folded those pamphlets. Camryn, Andrew, and Olive I am truly grateful for our friendship, even when you stole my money during Monopoly Deal. To Justine, thank you for being a rock for me throughout my time away and always keeping me updated on the town gossip. To Vanessa, I am so grateful for all the times we were able to yap and destress together, thank you. Mia, our fieldwork together really made me realize how much I love research and the people it allows you to bring into your life, and I am gratueful to have had that experience with you and our continued friendship. To the Šilovs fan club, thank you for keeping me entertained and entertaining my opinions throughout this degree.

Finally, to my family, Leo, Cathy and Mason McGuire. You have seen me through the lows and highs of this degree, and I am truly grateful for your support, and your pep talks when I needed it. Dad, I don't know if words can describe how much love I hold for you, and how much your support and words throughout my time away from home have allowed me to push forward. Mom, even though you can be slightly crazy sometimes, you will always beat any other mama bear, and I can't tell you how much it means to me to have that support in my corner, I love you. Finally, to Mason, I love you so much and am so grateful to have your support and to know that you are one call away when I need help procrastinating by having a chat about life.

Finally, thank you to Robbie, Lisa, and Tanya who were wonderful people to work, and thank you for welcoming and teaching so much for the short time we spent together. Also thank you to my committee members, Dr. Mark Johnson and Dr. Marysa Laguë for their feedback and insight. This research was generously funded through the Natural Sciences and Engineering Research Council (NSERC) Master's CGS-M and Discovery Programs, the Weston Family Foundation and Wildlife Conservation Society of Canada Boreal Research Fellowship, the Northern Scientist Training Program, and UBC Art's Graduate Student Research Award.

Chapter 1: Introduction

Inland lakes account for 5.6% ($1.44x10^6$ km^2) of the land surface area above 50°N, and are one of the most abundant freshwater ecosystems, accounting for 30% of the total area covered by wetlands, lakes, or rivers (Olefeldt et al., 2021). Lakes are important northern ecosystems, as they provide habitat for an array of organisms, support local communities by providing drinking water, provide a place for subsistence harvesting and resources for infrastructure building, and act as crucial carbon (C) managers at regional and global scales (Knopp et al., 2022). Their role cycling C is being impacted by the intensified warming occurring at northern latitudes, which has resulted in changes to water temperature, lake depth, and aquatic vegetation presence (Jansen et al., 2022; Mjelde et al., 2023). Currently, the influence of aquatic vegetation on C cycling is under-represented, particularly from methane (CH_4) research, despite aquatic vegetation potentially increasing CH_4 emissions by 1.3 to 1.8 times nearby open waters (Bodmer et al., 2024).

1.1 Aquatic CH_4 Cycling across Inland Waters

When discussing the influence of aquatic systems on C dynamics, there tends to be a focus towards ocean and coastal systems which account for a greater portion of global area. Inland aquatic systems have, historically, not received the same attention and were often disregarded as significant contributors to global C cycling because of their lower relative areal coverage (Cole et al., 2007). At regional scales, these systems can alter C balances significantly. For example, Cole et al. (2007) was one of the first studies to show that lakes specifically have higher rates of C storage per unit area than ocean or terrestrial systems, and can account for around 15% of carbon dioxide (CO_2) emissions from all inland waters. However, CO_2 represents only one portion of the inland waters C budget - CH_4 becomes a key factor in systems that have greater anoxic conditions and warm conditions, where in recent estimates the addition of CH_4 is shown to increase C outgassing estimates in inland waters (Bastviken et al., 2011; Drake et al., 2018). One ecosystem type that experience higher rates of anoxia and higher temperatures are small and shallow lakes, attributes that are common in Northern landscapes (Pi et al., 2022). While CH_4 is emitted at lower rates than CO_2 in most systems, it is a more potent GHG where it's emissions historically offset at least 25% of the estimated terrestrial GHG sink (Bastviken et al., 2011).

Methane emission in lakes typically occurs through three main transport pathways, diffusion, ebullition, and plant-mediated fluxes (Figure 1). The relative contributions of these pathways tend to vary based on a lake's abiotic and biotic characteristics (Bastviken et al., 2004; Deemer & Holgerson, 2021). Diffusion represents the transfer of gas through sediment-water-air interfaces and is the most commonly represented in lake CH_4 studies (Kuhn et al. 2021). Ebullition is gaining more representation in studies, but is still less commonly measured compared to diffusion, and describes the transport of gas through highly concentrated bubbles that are released from sediment (Bastviken et al., 2004). The final, and most understudied pathway, is plant-mediated transport through emergent or floating vegetation. This pathway represents the transfer of CH_4 from the production zones in the sediment to the atmosphere through plant tissues (Armstrong et al., 1996; Carmichael et al., 2014). Due to limited field measurements of plant-mediated transport from northern lakes and differences in methodologies, our understanding of the magnitude and drivers of plant-mediated CH_4 remain limited (Bodmer et al., 2024).

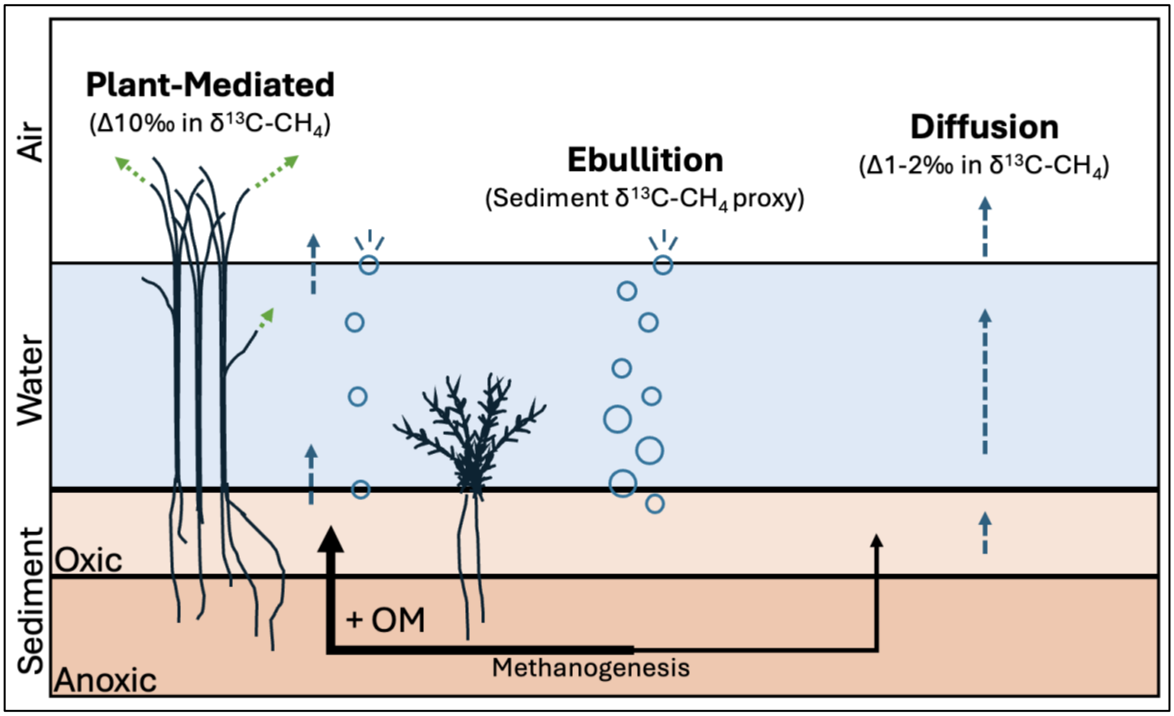

Figure 1 - Conceptual Diagram displaying CH_4 flux dynamics based on vegetation presence and plant functional group. Bubbles and blue dashed line represent ebullitive and diffusive flux respectively, with light green dashed arrows representing plant-mediated flux. Associated changes in $\delta^{13}C$ values between flux pathways and sediment values noted in parentheses. Relative potential sediment methanogenesis is represented through thickness of black arrows. Adapted from Carmichael et al. (2014), Parmentier et al. (2024), and Chanton (2005).

While there is limited knowledge surrounding plant-mediated flux, studies suggest plant-mediated transport has the highest potential for influencing total CH_4 emissions, contributing anywhere from 28 of 90% of total CH_4 flux (Carmichael et al., 2014). Most studies looking at the role of plant-mediated transport have occurred in temperate and southern boreal regions (Bodmer et al., 2024; Carmichael et al., 2014). It's possible that there are more studies from temperate and southern boreal zones due to more suitable conditions to support vegetation, but this may change due to shifts in vegetation distribution as the result of global climate change and anthropogenic influences (Alahuhta et al., 2011; Lobato-de Magalhães et al., 2023). For example, recent studies suggest vegetation is expanding into northern regions, with one key area being the Canadian shield

(Liu et al., 2025). As aquatic vegetation continues to expand, it is crucial that we understand how plant-mediated transport changes across regions, environmental conditions, and plant species.

1.1.1 Direct Influences of Aquatic Vegetation on the CH_4 Cycle

Aquatic vegetation can have direct and indirect influence on CH_4 production and emission. To quantify the direct effects of floating and emergent vegetation on emissions, the most common method is to deploy floating chambers directly on top of the exposed plant structures (Bastviken et al., 2011, 2023; Carmichael et al., 2014; Desrosiers et al., 2022). Aquatic vegetation is unique as they develop special aerenchymas tissues that transport the requisite oxygen (O_2) needed by plants through their shoots to roots in water-logged, anoxic environments (Striker, 2024; Villa et al., 2020). A by-product of this movement is the emission of gases in the opposite direction, where CH_4 and CO_2 move from sediments to the atmosphere. Typically, these species will have their shoots above the water, while their roots are in submerged sediments, creating what researchers have coined a "chimney effect." This effect describes the mechanism for which sediment-produced CH_4 can be directly transported by the plant by-passing oxic water columns that may consume 57-100% of released CH_4 (Armstrong, 1980; Bastviken et al., 2002; Saarela et al., 2019; Villa et al., 2020).

The amount of gas exchange occurring through the plant is dependent on whether the species has evolved to rely on humidity and pressure-driven gradients or concentration gradients. Commonly, vegetation studies have focused on species that employ the former dynamic, termed active bulk transport, such as common boreal species *Typha Latifolia* (Broadleaf Cattail; Desrosiers et al., 2022; Emilson et al., 2018), *Phragmites Australis* (Common Reed; Afreen et al., 2007; Kankaala et al., 2005) or *Equisetum fluviatile* (Water Horsetail; Kankaala & Bergström, 2004; Ojala et al., 2000). These species tend to have diurnal variations in CH_4 flux, where at peak times of solar insolation and temperatures they can move gases at higher rates and coincidentally these are the most common times for *in situ* CH_4 flux measurements (Chanton et al., 1993; Juutinen et al., 2004). The latter dynamic is what is employed by our study species, *Carex aquatilis* (water sedge), and others such as *Juncus effusus* (Common Rush; Henneberg et al., 2015) or *Peltandra virginica* (Green Arrow Arum; Whiting & Chanton, 1992). This gas exchange mechanism is known as molecular diffusion and is governed by the concentration gradient created through the movement of O_2 towards low-oxygen containing sediments which leads to the opposite movement of CH_4 towards the atmosphere (Vroom et al., 2022). Molecular diffusion has been said to be a slower gas exchange mechanism and thus a lower CH_4 emission mechanism, but few studies have tested this theory extensively through field measurement.

While emergent and floating plants are commonly studied, less attention has been focused on understanding the role of submerged vegetation in aquatic CH_4 cycling. Most existing research has used mesocosms or laboratory-based methods to address this knowledge gap (Hilt et al., 2022; Theus et al., 2023). From these studies, submerged vegetation has been shown to enhance CH_4 emission to varying degrees based on its density in the water column (Theus et al., 2023) or the respective lake temperature (Zhang et al., 2019), where higher densities and temperatures were related to higher CH_4 production from benthic vegetation. However, there is still some debate about whether this trend is universal, with some studies pointing to the fact that submerged vegetation may increase CH_4 oxidation in sediments, reducing the amount available for emission (Hilt et al., 2022; Liebner et al., 2011). Our understanding of CH_4 dynamics in the presence of submerged vegetation is relatively unconstrained, but through the support of laboratory and field measurements we can begin to understand how they alter aquatic CH_4 cycling.

1.1.2 Indirect Influences of Aquatic Vegetation

An indirect impact of aquatic vegetation on the CH_4 cycle includes changes to the biogeochemistry of lake sediments. Methane is produced in anoxic (i.e., lacking oxygen) sediments by CH_4 producing archaea (i.e., methanogens). However, a large portion of CH_4 produced in the sediments is consumed in oxygen rich surficial sediments or the water column through oxidation (Langenegger et al., 2022). The balance between CH_4 production and oxidation is especially important in littoral (i.e., shallow, lake) areas, due to warmer temperatures and a shallower water column, rich in oxygen (Hudspeth et al., 2024). These are also the regions of lakes where vegetation is often abundant. Recent studies have shown that vegetation can lead to the suppression of CH_4 production and increase oxidation of CH_4 in sediments (Bodmer et al., 2021; Hilt et al., 2022). This is facilitated through the transport of O_2 through the plants to the root system, where it can be expelled into the sediment, increasing O_2 availability in typically anoxic settings (Bodmer et al., 2021). However, these processes are still quite understudied in lakes, as the majority of knowledge surrounding plant-soil interactions and the CH_4 cycle are from wetland studies (Bodmer et al., 2021). While most of these processes are transferrable, there is still a need to understand how lakes function uniquely to other aquatic systems, especially when considering vegetation and its impact on the production, consumption, and transfer of CH_4 to the atmosphere.

Sediment CH_4 production is reliant on two key factors, temperature and organic matter (OM) lability. When either of these factors increase within a system, sediment CH_4 production has been shown to respond positively with large increases in potential CH_4 production (Duc et al., 2010; Grasset et al., 2019). Research recently has explored the connection between organic matter inputs and sediment CH_4 production, where they have shown that higher OM content and lability (i.e., decomposition of OM) leads to increased microbial CH_4 production (Emilson et al., 2018; Grasset et al., 2019). As well, when OM and temperature are paired together, studies have shown that sediments with more labile OM respond strongest to the temperature changes (Duc et al., 2010; Moras et al., 2024). The lability of OM can be tied to where it's sourced from, with aquatic, within-lake inputs being shown to support greater potential CH_4 production (Grasset et al., 2018). While this is recognized, many studies have concentrated on macro- and microalgae, alongside benthic (i.e., underwater) vascular plants. Benthic mosses on the other hand haven't received the same attention even though they are likely to be present along many lake shorelines and can provide the required OM to promote CH_4 production.

1.2 Research Objectives

The main objective of my thesis is to understand the magnitude and drivers of CH_4 flux along the vegetated and open water (i.e., unvegetated) areas of a shoreline of a mid-sized glacial lake within Kaska Dene ancestral territory, in now called northern British Columbia. My two main questions were (1) how does vegetation alter the magnitude of CH_4 flux within shallow lake zones? And (2) how would benthic vegetation presence and lake warming alter potential CH_4 production? To address my questions, I combined field observations of diffusive, ebullitive and plant-mediated fluxes, including measures of environmental variables, alongside an incubation to assess potential CH_4 production from sediments with and without benthic vegetation exposed to two temperature controls. The hypotheses supporting my research objectives, which are addressed throughout Chapter 2, include:

H1) CH_4 fluxes will be greatest in littoral areas with emergent vegetation rather than open water due to the ability for emergent vegetation to directly transport CH_4 through aerenchymas tissues, more effectively by-passing oxidation.

H2a) Incubations with benthic vegetation additions would have greater rates of potential CH_4 production compared to sediment only treatments due to more labile OM availability; and

H2b) Incubations exposed to higher temperatures will increase rates of potential CH_4 production due to promoted methanogenic activity under higher temperatures.

Chapter 2: Emergent water sedges (*C. aquatilis*) promote increased methane emissions and temperature sensitivity in littoral zones.

2.1 Introduction

Despite only accounting for around 5.6% of the terrestrial area above 50°N, freshwater lakes are estimated to emit 8-53% of natural methane (CH_4) from the Arctic-boreal region (Parmentier et al., 2024). While lakes are a potentially large source of natural CH_4 emissions (Rosentreter et al., 2021; Saunois et al., 2025), they are still underrepresented in comparison to more frequently studied, high emitting, wetlands (Kuhn et al., 2025; Li & Xue, 2023). Constraining high-latitude CH_4 emissions is crucial as emissions are expected to increase under accelerated atmospheric warming (Rantanen et al., 2022), due, in part, to increasing bottom water temperatures (Jansen et al., 2022; Kuhn et al., 2025) and northward shifts in aquatic vegetation (Emilson et al., 2018; Liu et al., 2025). These potential shifts in vegetation will likely have major repercussions for aquatic CH_4 cycling as plant-mediated emissions represent a dominant CH_4 transport pathway (Juutinen et al., 2003).

Plant-mediated transport of CH_4 is one of the three key pathways for CH_4 emission to the atmosphere (Bastviken et al., 2004). Along shallow shorelines, or littoral areas, ebullition (i.e., bubble emissions) is often the dominant transport pathway in many lakes (Wik et al., 2016), as potent CH_4 bubbles can by-pass consumption in the well oxygenated surface waters (Chanton, 2005). However, emergent vegetation can also increase CH_4 emission in littoral areas by transporting CH_4 from the sediment to the atmosphere through aerenchymas tissues, effectively by-passing the barrier that oxygenated sediment to water transitions present (Armstrong et al., 1996; Bodmer et al., 2024; Chanton & Whiting, 1996). In comparison to diffusive and ebullitive transport pathways, field-based measurements of plant-mediated emissions are vastly underrepresented, with recent flux synthesis studies, such as BAWLD-CH_4 (Kuhn et al., 2021) and ABCFlux v2 (Virkkala et al., 2025), excluding plant-mediated CH_4 measurements due to a lack of available data. While plant-mediated dynamics have been well studied in temperate lakes and wetland systems (Bodmer et al., 2024; Vroom et al., 2022), our understanding of these dynamics in northern lakes remains unconstrained (Kyzivat et al., 2022). In the Arctic-boreal region, only a handful of studies have quantified fluxes from emergent vegetation. These studies found that the presence of emergent vegetation increases CH_4 emission by 5 to 25% compared to open water littoral areas (Juutinen et al., 2003; Kyzivat et al., 2022; Liu et al., 2025). Elsewhere in temperate ecosystems, Bodmer et al. (2024), and references therein, have reported upwards of 30-80% higher emissions from emergent vegetation compared to adjacent open waters.

Emergent vegetation has been shown to directly and indirectly influence CH_4 production and emission. For example, as previously mentioned, emergent vegetation can directly impact CH_4 emission by facilitating the transport of CH_4 through aerenchymas tissues through pressure-mediated bulk transport or molecular diffusive mechanisms within the plant tissues effectively by-passing oxidation in the water column (Armstrong et al., 1996). Emergent plants can also indirectly alter CH_4 emission by providing labile organic matter (OM) substrates to sediments, enhancing the production of CH_4 (i.e., methanogenesis; Emilson et al., 2018; Grasset et al., 2019). However, the total impact of emergent vegetation on CH_4 emission remains relatively unconstrained due to a lack of extensive sampling temporally and spatially (Bastviken et al., 2023) and an unresolved understanding of how transport differs between emergent vegetation species (Kao-Kniffin et al., 2010).

In addition to emergent vegetation, changes in benthic vegetation are also likely to influence CH_4 dynamics in northern lakes. While some benthic vegetation species, like brown mosses, can promote CH_4 oxidation (Liebner et al., 2011), other studies suggest other benthic vegetation species increase CH_4 production by providing additional, labile OM substrates to the sediments, increasing methanogenesis (Bertolet et al., 2019; Grasset et al., 2018; Hershey et al., 2014), and, in some cases, leading to higher ebullitive emissions (Harpenslager et al., 2022). Benthic vegetation detritus has also been shown to increase sediment CH_4 production rates more than terrestrial inputs of OM due to their increased lability (Grasset et al., 2018). While these trends have been noted, there is still uncertainty surrounding the extent to which benthic vegetation detritus (i.e. dead benthic vegetation material), may alter CH_4 production and emission under a warming climate.

In our study we paired *in situ* measurements of CH_4 fluxes from all three transport pathways with laboratory incubations of potential sediment and benthic vegetation CH_4 production rates to identify the magnitude and drivers of CH_4 flux and temperature sensitivity from a northern boreal lake. We hypothesized that chamber-based CH_4 fluxes will be greatest in littoral sites with emergent vegetation compared to adjacent open water patches due to plant-mediated transport (H1). We also predicted that incubations with benthic vegetation additions would have greater rates of potential CH_4 production compared to sediment only treatments due to more labile OM availability (H2a). Additionally, potential CH_4 production will be more temperature sensitive in treatments with benthic vegetation than those with only lake sediments due to the compounding effects of greater OM and temperatures for methanogenesis (H2b).

2.2 Materials and Methods

2.2.1 Study Site Description

Wheeler Lake is in Kaska Dene traditional territory, in present-day Northern British Columbia (59.68937°, -129.16631°). It has a surface area of 2.4 km^2 with a maximum depth of around 25 m. The climate in the region is cold and temperate, with an annual average temperature of -1.6 °C and 711 mm of precipitation falling annually (*Dease Lake Climate: Weather Dease Lake & Temperature by Month*, n.d.). The watershed of Wheeler Lake is diverse, with coniferous forests dominating the watershed area, with small areas of mixed forest, shrublands, and wetlands (GeoBC, 2014).

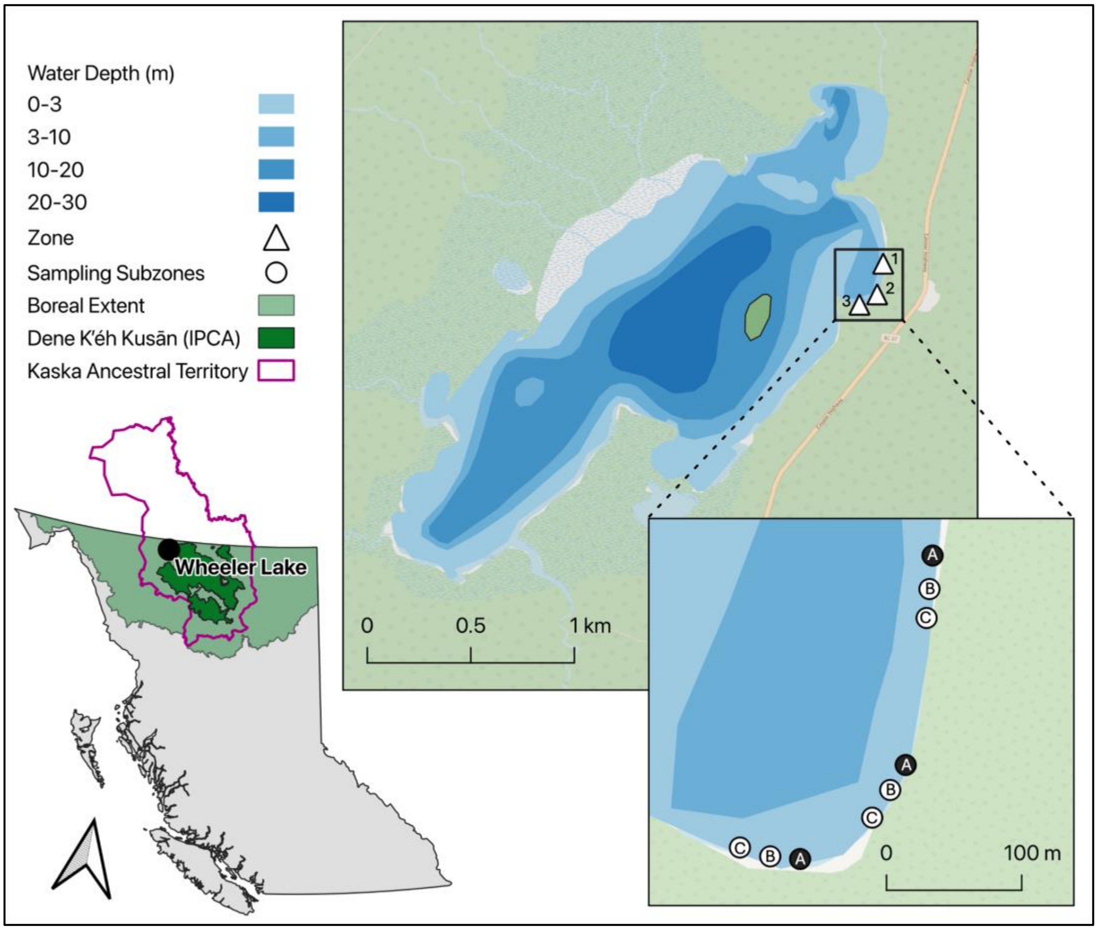

Figure 2 - Map of our study lake, Wheeler Lake, found within Kaska ancestral territory (purple) in present-day Northern British Columbia, Canada. Approximated water depth is shown in blue, with study zones (triangles) and sampling points (circles with replicate letters). The black circles represent the sampling points that had sensors where water and sediment temperature were collected. Map Layers were sourced from OpenStreetMap (2026), iNautical Depth Profiles (2026), BC Open Data Portal (2024,2025), World Wildlife Fund (2025).

Field data collection took place between July 14 to August 18th in 2025. We established three littoral sample zones along the northeastern shoreline, located ~100 m - 300 m away from a boat launch located north of Zone 1 (Fig. 2). Each zone contained three vegetated and three open sampling points (9 vegetation and 9 open water sampling points total), resulting in 18 unique sites. The dominant emergent vegetation across all zones was *Carex aquatilis* (water sedge; Fig. S1A). Benthic vegetation included *Scorpidium* (i.e., brown moss) family, *Potamogeton richardsonii*

(Richardson's pondweed), and *Hippuris vulgaris* (common mare's tail). Zones 1 and 3 had the lowest visual coverage of benthic vegetation, consisting of sparse-moderate amounts of vascular benthic vegetation (i.e., pond weed) and some mosses. Zone 2 had the densest benthic coverage, with thick mats of mosses covering most of the area and pond weed or mare's tail occupying the remaining patches (See Fig. S1B).

2.2.2 Water Chemistry and Environmental Variables

Near the beginning (July 27), middle (August 8), and end of the field season (August 18), surface water samples from each vegetated and open zones (n = 18) were collected from ~5-10 cm depths below the surface using a sterilized 50mL Centrifuge Tubes (Celltreat Scientific Products). Samples were kept cold for 2-6 hours until they were frozen until analysis (~2-3 weeks), following the Natural Resources Analytical Laboratory (NRAL, University of Alberta) sampling protocols. Dissolved Organic Carbon (DOC), Total Organic Carbon (TOC), and Dissolved/Total Nitrogen (TN) were analyzed using a TOC-L carbon analyzer with an ASI-L and TNM-L module after samples were filtered at 0.45 mm (Shimadzu). Dissolved ions were analyzed using a Thermo iCAP6300 Duo ICP-OES equipped with an autosampler. We determined Wheeler's Trophic State Index using a combination of TN and PO_4^{3-}-P values and Secchi depth at the deepest point of the lake following Carlson (1977) and Kratzer and Brezonik (1981).

We measured dissolved oxygen (DO), specific conductivity, pH, total dissolved solids (TDS), and surface water temperature at 5-10 cm below the water surface during each flux measurement (Yellow Springs Instruments 550A 50' DO Meter; Hanna Probe HI98195). We measured surface and sediment temperatures every 2 hours over the 35-day sampling period at one open water and one vegetated site in each zone using pendant sensors (UA-001-64; UA-002-64 Onset HOBO; see Fig. 2 black dots). Prior to use in statistical analysis, we averaged the temperatures at daily, three-day and seven-day intervals, with a moving average being applied to the latter two averages. We used three-day averages during our analysis as it was consistently the best AIC performer, and this time span may help to account for a lag in methanogenic temperature responses.

In open and vegetated sampling points, we collected surface water samples for dissolved concentrations of CH_4. Following the headspace equilibration method, we pre-rinsed 10 mL syringes with lake water and collected 5 mL of sample from 5-10 cm below the surface. We then added 5 mL of air before shaking the syringe vigorously for 2 minutes (Noyce et al., 2014). We slowly ejected the water by inverting the syringe and then kept the remaining 5 mL air sample stored within the syringe using an air-tight three-way stopcock until the sample could be analyzed within 12-48 hours. Samples were analyzed for CH_4 using a LI-7810 Trace Gas Analyzer (LICOR, Lincoln, United States) equipped with a single-inject loop. Two mL of each sample were injected into the single-sample processing loop, pre- and post-injection concentrations noted and corrected for the final concentration following LICOR single-inject protocols using the effective volume of the closed loop and a known standard of CH_4 (Xu, 2020). CH_4 partial pressures were converted to dissolved CH_4 concentrations using the neonDissGas package in R (v4.5.2; Cawley, 2007; R Core Team, 2025).

2.2.3 Diffusive and Plant-Mediated Chamber Fluxes

We sampled diffusive and plant-mediated fluxes using opaque floating chambers. We measured fluxes from each of the nine sampling points seven times over 12 different days throughout the field season (i.e., all three zones could not logistically be sampled on one day). Floating chambers (polyethylene; V = 9.9 and 15 L for open water and vegetation measurements, respectively) were equipped with a small fan to ensure well-mixed air within the chamber headspace. Each chamber was wrapped in reflective tape to prevent excessive heating and was equipped with a removable septum to aid in equilibrating pressure differences when placed on the water surface. Each chamber was fitted with two gas-tight tubes equipped with stopcocks for manual extraction of air samples from the chamber headspace (see Fig. S2a/b).

Chambers were placed on the water surface for 30 minutes, with periodic sampling of the headspace at 0, 15, and 30 minutes (Kankaala et al., 2005). Gas samples were extracted using 60 mL polypropylene syringes (MedNeedles, Ontario, Canada). For the first two sampling points (0 and 15 minutes), the samples were stored within the sampling syringe with a closed stopcock and were analyzed on the LICOR within 24-36 hours following the same single-inject methods outlined above. The final sample (30 minutes) was stored in a N_2-flushed, 20 mL gas vial (Thermo Scientific) sealed with butyl rubber septum to allow for immediate analysis and long-term storage.

Chamber fluxes (i.e., diffusive + plant-mediated) were calculated following DelSontro et al. (2016). We first calculated the change in CH_4 concentration (i.e. slope) across the 30-minute sampling period (i.e., 0, 15, 30 minutes) and filtered out any ebullitive events based on manual inspection of irregular, concentration peaks across the time period (n = 3; Bastviken et al., 2004). Fluxes were then determined following the ideal gas law:

$$(1)\ Flux = \frac{(s \times P_{atm})}{RT} \times \frac{V_{ch}}{A_{ch}} \times 12.01 \times 86.4$$

Where s is representative of the slope of concentration accumulation in the chamber (ppm s^{-1}), P_{atm} (kPa) is the atmospheric pressure at the time of sampling, RT is the result of the ideal gas law (R) and temperature (K) at the time of measurement, V_{ch} and A_{ch} are the volume (m^3) and area (m^2) of the chambers, and finally to convert from µmol m^{-2} s^{-1} to mg C-CH_4 m^{-2} d^{-1}, we used a conversion factor of 86.4 to go from µmol m^{-2} s^{-1} to mmol m^{-2} d^{-1}, then 12.01 for the final conversion to mg C-CH_4 m^{-2} d^{-1}. Before any further calculations, we removed any remaining extreme outliers that were found beyond the range of 3 times the interquartile (IQR) range of open water and vegetated areas, respectively (n = 3). To differentiate between diffusive and plant-mediated flux within the vegetated chamber measurements, we assumed that the open water measurement of diffusive flux was equivalent to that found in our vegetated sample area and subtracted the flux found in the paired open water flux from the total chamber flux value following the methods outlined in Desrosiers et al. (2022). If either vegetated or open water fluxes were removed for outliers, the median flux from the proper zones were applied in the outlier's place.

2.2.4 Ebullitive Flux

To measure ebullitive fluxes we used inverted funnels ("bubble traps", polyethylene, area = 0.059 m^2; Fig. S2c; Kuhn et al., 2023) installed in vegetated and adjacent open water sampling points (three pairs per zone; 18 traps total). Bubble traps were installed at the beginning of the field season (July 14th) and were sampled from an inflatable kayak 14 times over 34 days, with sampling occurring every one to nine days. The total accumulated air within bubble traps was marked each sampling day, and if it there was enough air (i.e., >1mL), we would manually pull samples from bubble traps using syringes and store it within nitrogen (N_2) flushed, 20 mL gas vials (Thermos Scientific) sealed with butyl rubber septum. In total, we collected 228 measurements of accumulated bubble volume, and 113 concentration samples for analysis. To determine the concentration of CH_4, samples were analyzed within four months following the same single-inject methods outlined above. Prior to analysis, samples were diluted using a 1 to 400 mL ratio of sample to N_2 gas to achieve concentrations that fell below the upper range for the LICOR (100 ppm). To calculate ebullitive fluxes, we followed the methods outlined by Wik et al. (2013).

2.2.5 $\delta^{13}C$-CH_4 Isotope Analyses

To examine potential CH_4 transport and production/consumption pathways we analyzed a subsample of paired floating chamber and bubble flux samples for the stable carbon isotopic composition of CH_4 ($\delta^{13}C$-CH_4). The sub-samples represented measurements from the beginning, middle, and end of the field season across sampling points (Table S1). Samples were run on a Picarro G2201-i Isotopic Analyzer. Floating chamber samples were run on the iCO_2-iCH_4 High Precision mode, while ebullitive samples were analyzed on the iCO_2-iCH_4 High Range Reading mode to account for higher concentrations. Bubble samples were diluted with ultra zero air prior to analysis to within the instrument dynamic range (< 100 ppm CH_4). Reference standards with $\delta^{13}C$-CH_4 of -26.5‰, -45.5‰, and -68.6‰ were run alongside samples to correct for instrument accuracy and drift. After excluding samples that did not have high enough CH_4 concentrations for the isotopic analysis (< 1.8 ppm) and one vegetated and open water chamber pair that had an outlying $\delta^{13}C$-CH_4 value above the rest of the vegetated chamber samples, we had 10 paired floating chamber samples (20 total) and 17 bubble trap samples (8 vegetation and 9 open water).

2.2.6 Emergent Vegetation Biomass and Characterization

We measured the diameter and heights of *Carex aquatilis* stands to get a representative estimate of plant biomass at the end of the sampling season. At four sampling sites across the three zones, 20-30 shoots of vegetation were measured for diameter and height, cut, and then stored in paper bags for biomass calculations. To calculate the dry weight of these samples, cuttings were dried at 60°C for 48 hours until the dry weight (grams dry weight (g DW)) stabilized (Kankaala et al., 2005). To obtain a final biomass calculation, the dry weights were normalized over the number of shoots and then over the area cut, which approximated the chamber footprint. This value was then applied to the counts of shoots from each vegetated chamber flux, found using methods described below, resulting in an estimated DW biomass per metre squared (g DW m^{-2}).

To characterize the presence of vegetation for each chamber measurement, a top and side view photo of the vegetation was captured before each vegetated flux measurement (see Fig. S3). These photos were then used to classify the number of shoots in the chamber area, the percentage of surface area covered by the vegetation, and finally greenness indices, including the Greenness Chromatic Coordinate and Excessive Greenness (Nijland et al., 2014). We counted shoots manually, while the latter two measures were calculated using RGB band information pulled from ImageJ software (Schneider et al., 2012; Example workflow in Fig. S3c).

2.2.7 Benthic Vegetation and Sediment Potential Methane Incubation

We used *ex-situ* incubations to calculate potential CH_4 production rates from littoral sediment and benthic vegetation. The samples were collected within the littoral area from the upper 3-5 cm of the sediment and water columns, respectively. We collected lake water using pre-rinsed 1 L amber bottles. Water was kept refrigerated until the incubation. Using sterile gloves, we manually collected sediment samples from a shallow area just outside of the open water area in Zone 3 due to ease of accessibility. Sediments were stored within air-tight plastic bags with all air bubbles removed. We collected a representative mixture of benthic mosses from the thick mat of vegetation found in Zone 2 and stored the samples in air-tight plastic bags. Benthic vegetation and sediment materials were frozen at -20 °C until the incubation.

We included three treatments in the incubation including Sediment Only, Sediment + Benthic vegetation, and Benthic Vegetation only (Fig. 3). Each treatment had five replicate vials with 20 mL of lake water in each vial and 10 g of sediment or benthic material or a combination of both (i.e., 5 g sediment + 5g benthic vegetation). We flushed all vials pre-incubation with N_2

for 15-20 minutes to ensure anaerobic conditions and minimal CH_4 in the vial headspace. The treatments were exposed to two temperatures, 20°C and 30°C, which reflects approximate average sediment temperatures (see Fig. S4), and a scenario of potential warming of 10°C, which allowed for Q_{10} analysis.

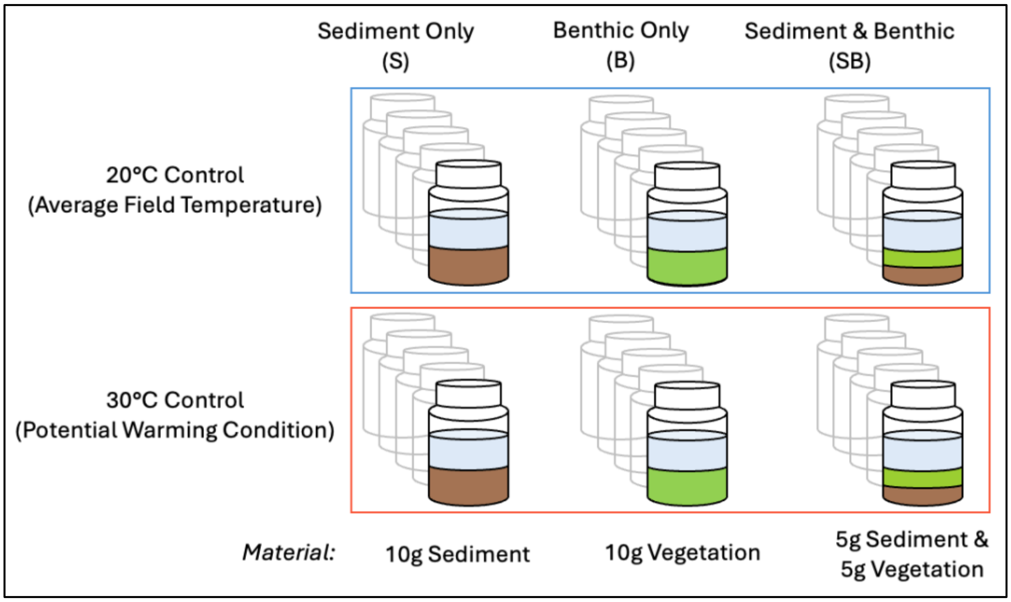

Figure 3 - Potential CH_4 Production incubation experimental scheme. The treatments (left to right), with material distributions underneath, which includes a Sediment Only (S) treatment, Benthic Only (B), and finally Sediment and Benthic vegetation (SB). Treatments were further split into two temperature controls reflecting the average field (top) and potential warming scenario (bottom) sediment temperatures. There were 5 replicates per treatment, per temperature control, reflected by grayed out vials.

After flushing the headspaces, all vials were wrapped in aluminum foil and were pre-incubated in the dark for 10 days in the CONVIRON Plant Growth Chambers (Model A1000) at their respective incubation temperatures to allow for each treatment to equilibrate to their respective environment. After 10 days, all vials were shaken, flushed with N_2 to minimize CH_4 in the headspace, before we pulled four mL of headspace gas to establish the baseline concentration in each vial (day 0). The vials were placed back into the growth chamber and then sampled for headspace pressure and concentration on days 1, 2, 3, 7, and 10. To sample the headspace, we first injected two mL of N_2 in the headspace, prior to pulling out the sample. All headspace samples were analyzed for CH_4 concentrations (ppm) on the sampling day using the LICOR single-inject methods described earlier, but with dilution factors of 1:100 to 1:1000, to fit the upper limit of the analyzer. We converted from ppm CH_4 to μg by incorporating the headspace volume, temperature, and pressure following the ideal gas law (Hodgkins et al., 2014).

We calculated potential CH_4 production rates by determining the change in CH_4 concentration between each timestep. We then normalized these rates by dry weight (g) of the sediment/benthic material. We quality controlled the data by removing the highest and lowest production rates from each treatment on each sampling day, which resulted in three measurements per each day and treatment, before calculating daily average production CH_4 rates. Finally, we quantified temperature sensitivity of potential CH_4 production across the treatments by calculating Q_{10} rates following Duc et al. (2010).

At the end of the sampling period, we measured volumetric headspace (mL) before placing the vials within a drying oven at 50°C for one week to dry the sediments and benthic vegetation. Two replicates of dried samples from each treatment were analyzed for total C and N by grinding the dry samples prior to analysis on a CHN elemental analyzer (Vario Micro Cube, Elementar, Germany). To determine OM content from each treatment we used the loss on ignition method (LOI; Heiri et al., 2001; Karvinen et al., 2015). Samples were ground, placed into crucibles, and weighed before being placed in a muffle furnace at 500°C for 7 hours. Crucibles were cooled overnight and then weighed in the morning to obtain a final weight to calculate the total amount of OM burnt off during the process.

2.2.8 Statistical Analyses

All statistical analyses were conducted with R statistical software (v4.5.2; R Core Team, 2025). We use generalized linear mixed models (GLMMs; glmmTMB(); Brooks et al., 2026) to examine correlations between daily-averaged flux across open and vegetated sites, zones, and environmental variables. We chose to use GLMMs as our data included a nested structure with repeated measures alongside non-normal distributions, numerous zeroes, and non-independence factors (Zuur et al., 2010). We accounted for these data distribution characteristics by log-transforming our data, and using gamma, or zero-inflated gamma, distributions with a log link function (Al-Haj et al., 2022). For comparisons between vegetated and open water environmental response variables, we used GLMMs with vegetation/open water and zone as fixed effects and site as the random effect to account for repeat sampling. To identify correlations between environmental variables and CH_4 flux, we used the same model set-up but replaced vegetation presence with the parameter of note (see Table S2 for model outputs). We conducted post-hoc pairwise Estimated Marginal Means (emmeans(); Lenth et al., 2026) tests to reveal whether parameters differed significantly across open water and vegetated sites, across zones, and with their interaction. We also tested differences in potential CH_4 production across sampling days, treatments, and temperature controls, with various interactions, using GLMMs with interactions between treatments and sampling days and replicates as random effects (see Table S3). We used emmeans() again to reveal when treatments differed significantly from each other within and across temperature controls on individual sampling days and across the sampling period. All models were run against a significance level of 5% (alpha =0.05).

2.3 Results

2.3.1 Lake Chemistry and Temperatures

Water chemistry for Wheeler Lake indicates that it is oligo-mesotrophic, with a trophic state index of 31, Secchi depth of 7 m, and relatively low concentrations of DOC (6.00 ± 1.22 mg L^{-1}; mean ± std), TN (0.35 ± 0.06 mg L^{-1}), and phosphate (PO_4^{3-}; 4.74 ± 2.55 µg L^{-1}; See Table 1 for full suite of water chemistry variables). There were no significant differences between surface water chemistry parameters from open water and vegetated sites like surface water pH (8.20 ± 0.44; n = 120), DO (109 ± 8.33 %; n = 120), or conductivity (888 ± 10.4 mS cm^{-1}; n = 120). Three-day averaged temperatures, including sediment (15.8 - 20.6°C; "min - max"), bottom water (16.4 - 21.8°C), and surface water (16.3 - 22.2°C) also did not statistically differ between open and vegetated areas. Of the suite of dissolved ions and nutrients, TN and ammonium (NH_4-N) were the only analytes to significantly differ, both with lower concentrations in open water (0.31 ± 0.02 mg L^{-1} and 14.5 ± 3.18 µg L^{-1}; n = 9) compared to vegetated areas (0.38 ± 0.07 mg L^{-1} and 19.2 ± 8.44 µg L^{-1}; n = 9; p = 0.0002; emmeans). Finally, dissolved CH_4 concentrations trended towards being higher in vegetated areas (473 ± 199 nmol L^{-1}) compared to paired open water areas (428 ± 228 nmol L^{-1}; n = 119, p = 0.12; emmeans), but this difference was not statistically significant.

Table 1 - Biogeochemical characteristics of Wheeler Lake split by Open water and Vegetated Values. Data was sourced from field collection over July and August 2025. Means are given along with the standard deviation.

Physical and Biogeochemical Characteristics		
	Open Water	**Vegetated**
pH *(n = 126)*	8.19 ± 0.44	8.20 ± 0.44
Conductivity *(n = 126)*	889 ± 8.59 mS cm^{-1}	887 ± 12.0 mS cm^{-1}
Total Dissolved Solids *(n = 126)*	444 ± 4.30 mg L^{-1}	444 ± 5.48 mg L^{-1}
Dissolved Oxygen *(n = 126)*	109.6 ± 8.48 %	107.4 ± 8.09 %
Dissolved Organic Carbon *(n = 18)*	5.64 ± 0.78 mg L^{-1}	6.36 ± 1.50 mg L^{-1}
Total Carbon *(n = 18)*	44.2 ± 2.15 mg L^{-1}	45.1 ± 2.51 mg L^{-1}
Total Organic Carbon *(n = 18)*	7.30 ± 1.72 mg L^{-1}	8.03 ± 0.44 mg L^{-1}
Total Inorganic Carbon *(n = 18)*	36.9 ± 1.48 mg L^{-1}	37.0 ± 2.47 mg L^{-1}
Total Nitrogen *(n = 18)*	0.31 ± 0.02 mg L^{-1}	0.38 ± 0.07 mg L^{-1}
NO_3-N *(n = 18)*	0.02 ± 0.006 mg L^{-1}	0.03 ± 0.02 mg L^{-1}
NH_4-N *(n = 18)*	14.5 ± 3.18 µg L^{-1}	19.2 ± 8.43 µg L^{-1}
PO_4-P (*n = 18*)	4.00 ± 0.59 µg L^{-1}	5.49 ± 3.49 µg L^{-1}
Fluoride *(n = 18)*	0.12 ± 0.01 mg L^{-1}	0.14 ± 0.02 mg L^{-1}
Chloride *(n = 18)*	0.40 ± 0.05 mg L^{-1}	0.39 ± 0.01 mg L^{-1}
SO_4-S *(n = 18)*	13.2 ± 0.73 mg L^{-1}	13.5 ± 0.33 mg L^{-1}
3-day Averaged Surface Water Temperatures *(n = 24)*	19.1 ± 1.66 °C	19.2 ± 1.68 °C
3-day Averaged Bottom Water Temperatures *(n = 24)*	18.9 ± 1.51 °C	18.9 ± 1.54 °C
3-day Averaged Sediment Temperatures *(n = 24)*	18.4 ± 1.23 °C	18.1 ± 1.35 °C

2.3.2 Plant Biomass and Greenness

The mean number of *Carex* shoots and their biomass across sampling plots was 24 ± 8 and 94.9 ± 30.9 g DW m^{-2} respectively. The most abundant average vegetation was found in Zone 3 which had 27 ± 3 shoots and a biomass of 108.9 ± 32.4 g DW m^{-2} which is similar to Zone 2 (25 ± 8 shoots and a biomass of 98.0 ± 29.9 g DW m^{-2}). Zone 1 had statistically less vegetation than zone 3, but not Zone 2 (shoot count = 19 ± 5 and mean biomass of 77.5 ± 21.4 g DW m^{-2}; emmeans;

$p = 0.001$). Green Chromatic Coordinate (GCC) was similar across all three zones, with the highest GCC reported from Zone 2 (0.37 ± 0.02), followed by Zone 1 (0.36 ± 0.01) then Zone 3 (0.36 ± 0.01). Temporally, GCC generally declined during the sampling period that ranged from mid-July to the end of August (Conditional $R^2 = 0.44$, p-value < 0.001; Fig. S5).

2.3.3 Methane Fluxes

The mean chamber CH_4 flux between open (diffusive) and vegetated (diffusive + plant-mediated) sites significantly differed (emmeans; p-value < 0.0001), with lower fluxes from open water (22.7 ± 19.5 mg C m^{-2} d^{-1}; n = 60) compared to vegetated areas (86.2 ± 57.3 mg C m^{-2} d^{-1}; n = 61; Fig. 4a). Bubble flux, representing the volume of bubbles collected (Fig. S6), was higher from open (0.28 ± 0.34 L m^{-2} d^{-1}; n = 103) versus vegetated areas (0.20 ± 0.25 L m^{-2} d^{-1}; n = 99; emmeans; p-value = 0.02). However, CH_4 ebullition did not differ between open water (22.0 ± 28.0 mg C m^{-2} d^{-1}) and vegetated areas (18.1 ± 28.7 mg C m^{-2} d^{-1}**;** emmeans; p-value = 0.13; Fig. 4b). There were no significant temporal patterns across the sampling period for any flux (Table S2).

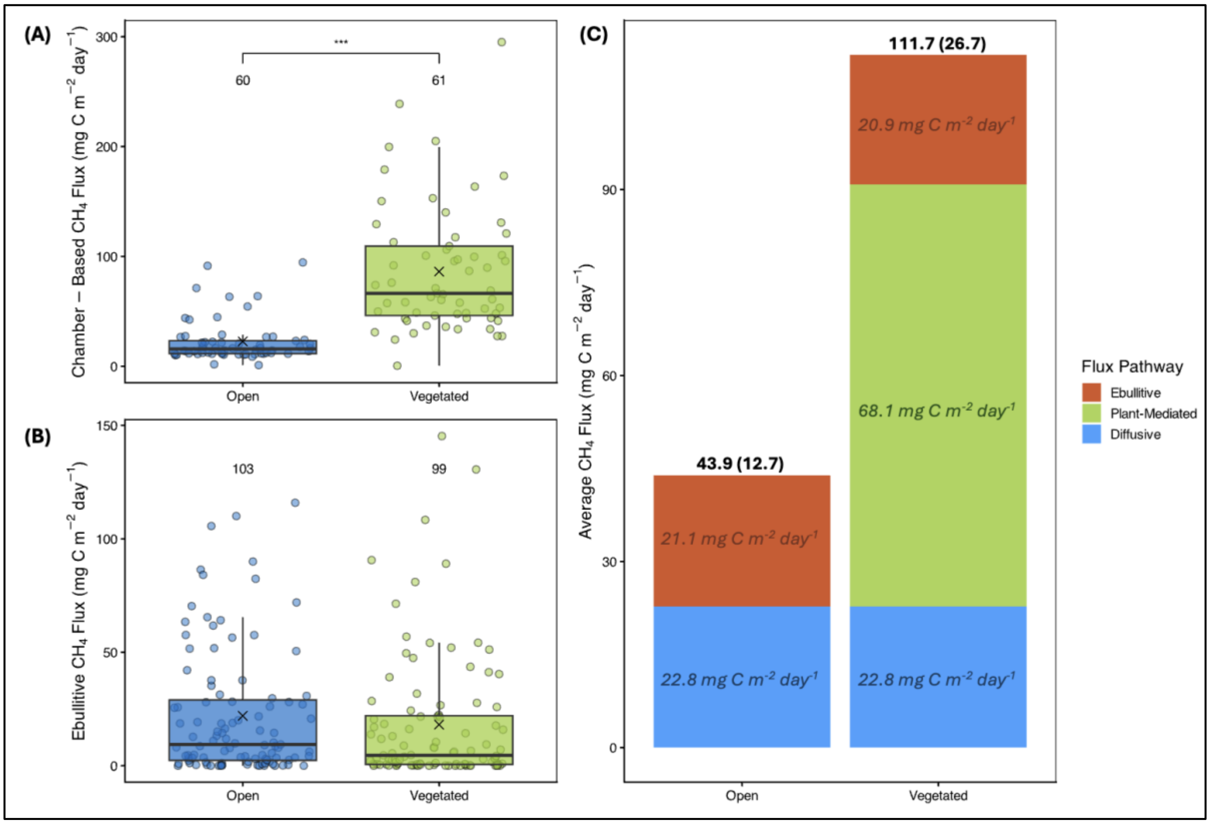

Figure 4 - Direct plant-mediated flux contributes disproportionate amounts of CH_4 across littoral areas. (A) Diffusive and plant-mediated CH_4 flux and (B) Ebullitive flux from open and vegetated sampling points. Boxplots represent the median and first and third quartile ranges, with the cross representing the mean. (C) Relative contribution of flux pathways to overall flux in open water and vegetated areas. Numbers within pathway bars represent the average flux from each pathway, with total flux at the top with the standard deviation in parentheses.

Total flux (ebullition + chamber flux) was greatest in vegetated areas (111.7 ± 26.7 mg C m^{-2} d^{-1}) compared to open waters (43.9 ± 12.7 mg C m^{-2} d^{-1}; Fig. 4c). While we assumed our measured diffusive emissions were the same across vegetated and open water zones (22.8 ± 6.36 mg C m^{-2} d^{-1}), diffusion had a larger contribution to open water fluxes, contribution 52% of total flux, compared to only 20% in vegetated plots. Plant-mediated fluxes have the highest contribution across pathways at 61% (64.6 ± 18.99 mg C m^{-2} d^{-1}) of the total flux. Ebullition in both open water and vegetated areas contribute the least to overall flux, with ~48% (21.1 ± 15.2 mg C m^{-2} d^{-1}) in open water and 19% (20.9 ± 13.9 mg C m^{-2} d^{-1}) in vegetated areas (Fig. 4c).

2.3.4 $\delta^{13}C$-CH_4 Values

Methane isotopic composition ($\delta^{13}C$-CH_4) did not differ between open water and vegetated sites (Fig. 5) for chamber measurements (-66.9 ± 11.4 and -71.8 ± 7.3‰, respectively; emmeans; $p = 0.09$) nor for ebullitive measurements (-60.2 ± 6.0 and -56.2 ± 5.5‰; emmeans; $p = 0.31$). Chamber and ebullitive $\delta^{13}C$-CH_4 values for open water areas also did not differ (emmeans; $p = 0.22$). However, chamber $\delta^{13}C$-CH_4 values from vegetated plots were significantly lower than open water and vegetated ebullitive values (emmeans; $p < 0.04$), with the largest difference between vegetated chamber and ebullitive measurements (difference = -16‰; emmeans; $p = 0.002$).

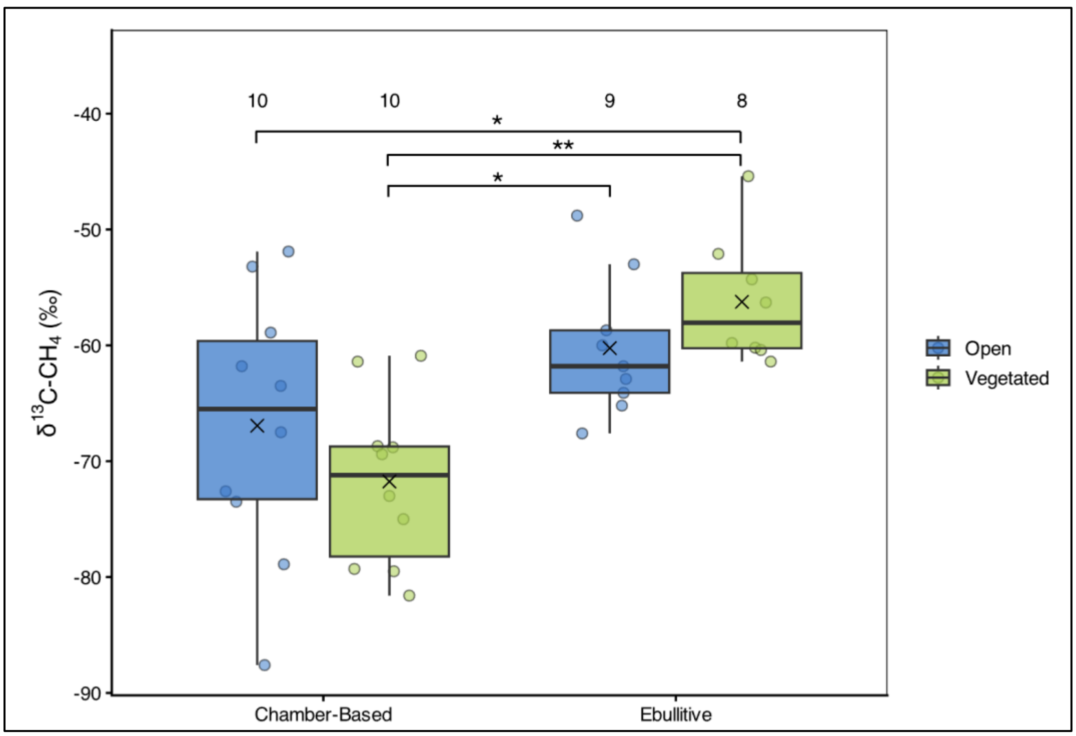

Figure 5 - $\delta^{13}C$-CH_4 values likely indicate more efficient transport through plants. Values are split into open and vegetated samples and grouped by measurement technique. Means are represented by crosses, with significant comparisons based on emmeans pairwise post-hoc tests represented by asterisks (* = 0.05, ** = 0.01).

2.3.5 Influence of Environmental Factors on CH_4 Fluxes

Using GLMMs, we found few significant correlations between environmental variables and diffusive, ebullitive or plant-mediated fluxes from vegetated or open water sites (Table S3). For example, sediment temperature, water temperature, pH, DO, biomass, and greenness indices did not significantly explain any variation in flux. The only significant trend we found was a negative correlation between depth and ebullitive flux in open water areas (GLMM; $p = 0.044$; Marginal $R^2 = 0.13$). We do observe non-significant trends, where relationships between depth, sediment temperature and plant-specific metrics are noted (Fig. 6). We found variable trends with depth depending on the specific flux (Fig.6A-C), where we saw negative trends for depth with plant-mediated flux and vegetated chamber fluxes, but a positive trend for vegetated ebullitive flux and open water chamber flux. As for sediment temperature (Fig.6D-F), we see neutral to slightly negative trends across plant-mediated and vegetated chamber and ebullitive fluxes, while open water chamber flux and ebullition show positive correlations with depth. Finally, plant-specific variables such as biomass (Fig. 6G), percent cover (Fig. 6H), and GCC (Fig. 6I) all have neutral to slightly positive relationships.

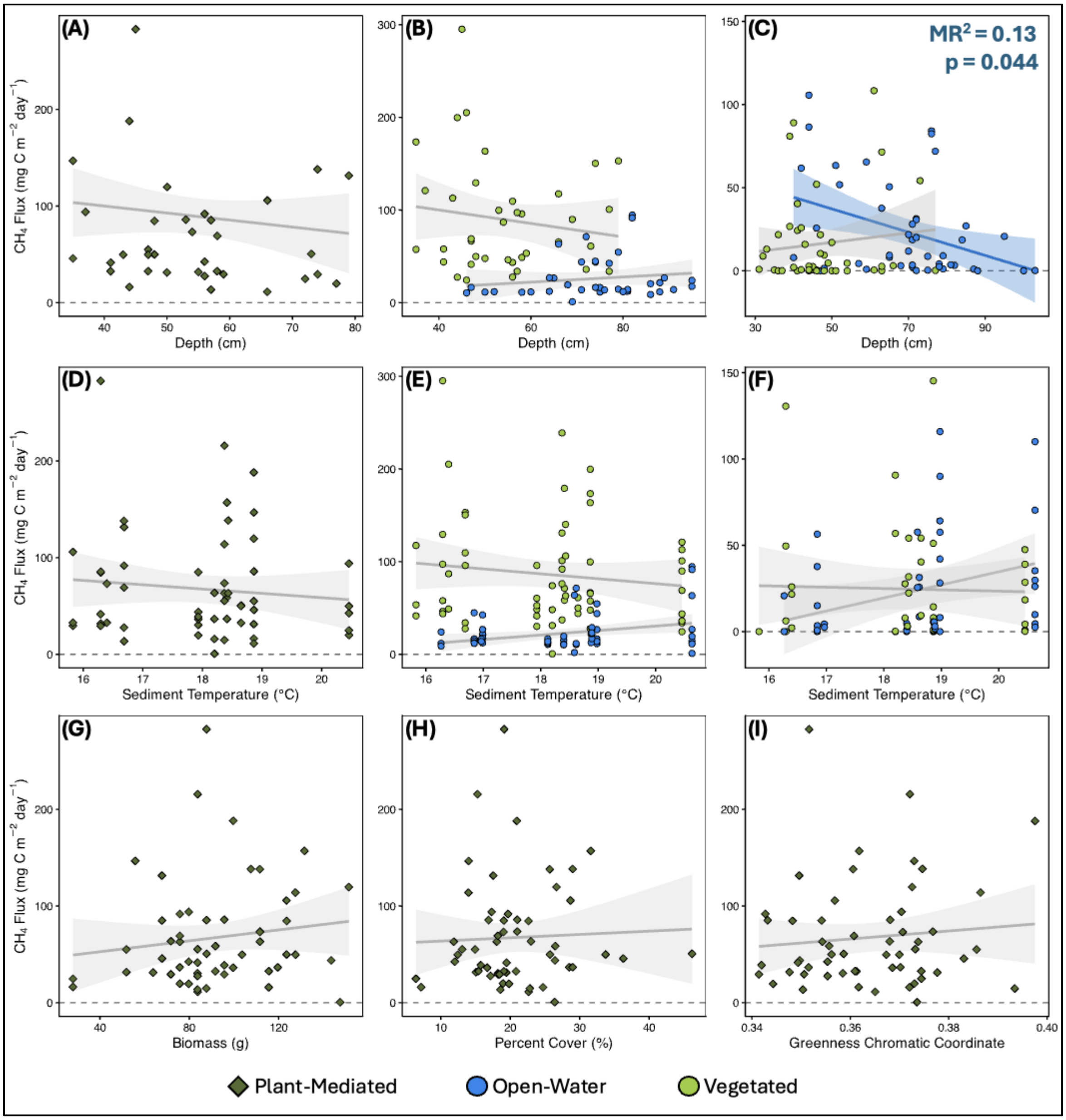

Figure 6 – Environmental variables were not strong predictors of CH_4 flux. Each row represents individual parameters modelled against plant-mediated (left; dark green diamonds), chamber-based (centre column), and ebullitive flux (right column). Points represent individual flux measurements, and the grey lines and shading represent the modelled relationships.

2.3.6 Potential Methane Production from Sediment and Benthic Vegetation Incubations

Across the 10-day incubation there was a non-linear increase in potential CH_4 production across treatments and temperature controls (Fig. 7; Table S3). The average potential CH_4 production rates at 20°C ranged between 0.16 nmol CH_4 g DW d^{-1} (sediment only, day 2) and 46.6 nmol CH_4 g DW d^{-1} (benthic only, day 10). In comparison, production rates in the 30°C treatments ranged from 0.52 nmol CH_4 g DW d^{-1} (sediment only, day 1) to 221 nmol CH_4 g DW d^{-1} (benthic only, day 10).

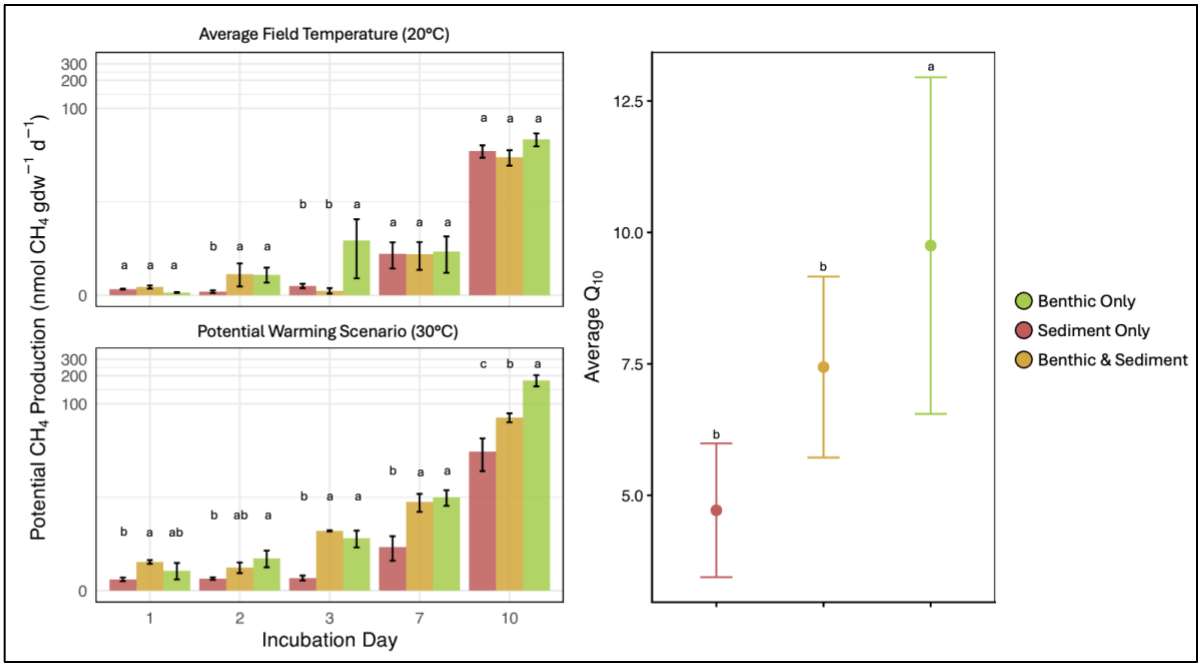

Figure 7 - The combination of higher temperature and OM input drives potential CH_4 production. (A) Progression of potential CH_4 production rates across a two-week incubation, split by 20°C (top) and 30°C (bottom) temperature controls. Potential production rates are shown on a pseudo-log transformed for easier comparison and standard error bars based on treatment averages are shown on top of each bar. (B) Averaged Q_{10} values by treatment, with standard error bars. Letters denote significant differences in treatment responses.

Across the incubation sampling period, the Benthic Only and Benthic + Sediment treatments generally had higher production rates than the Sediment Only treatment and these significances were strengthened at 30°C compared to 20°C. On days 1, 2 and 3, the Sediment Only treatment had the lowest production rates, with an average production rate of 0.31 ± 0.14 and 0.60 ± 0.03 nmol CH_4 g DW d^{-1} for the 20°C and 30°C controls, respectively. At 20°C, the Sediment + Benthic treatment had similar production rates to the Sediment Only treatment on days 1-3 (average = 0.57 ± 0.45; Table S5 & S6), but at 30°C, Sediment + Benthic production rates more closely resembled the Benthic Only treatment (average = 2.3 ± 1.6 and 2.1 ± 1.2 nmol CH_4 g DW d^{-1}, respectively; Table S4 & S5). At 20°C, Benthic Only treatments, which had an average production rate of 1.0 ± 1.7 nmol CH_4 g DW d^{-1} from days 1 to 3, significantly differed from Sediment Only on days 2 and 3 (emmeans; $p < 0.004$), but only from Sediment + Benthic on day 3 (emmeans; $p < 0.0001$). On days 7 and 10, there were no significant differences at 20°C, where Sediment Only (18.5 ± 8.0 nmol CH_4 g DW d^{-1}), Sediment + Benthic (19.5 ± 22.9 nmol CH_4 g DW d^{-1}), and Benthic Only treatments (24.7 ± 19.4 nmol CH_4 g DW d^{-1}) had similar ranges. However, on days 7 and 10 at 30°C, Sediment Only treatments had significantly lower production (2.92 and 31.5 nmol CH_4 g DW d^{-1}, respectively) than the Sediment + Benthic (11.3 and 93.4 nmol CH_4 g DW d^{-1}, respectively) and Benthic Only treatments (26.8 and 221 nmol CH_4 g DW d^{-1}, respectively). Finally, the Benthic Only treatment had the highest Q_{10} value of 9.8 ± 7.2, followed by Sediment + Benthic (7.4 ± 3.4), and then the Sediment Only treatment with a Q_{10} of 4.7 ± 2.8, which was significantly lower than Benthic Only (emmeans; $p = 0.03$).

The Benthic Only treatments had the highest average OM content (50.8 ± 3.18%), followed by Sediment + Benthic (12.1 ± 0.36%), and then Sediment Only (4.31 ± 0.07%), all of which were significantly different from each other (emmeans; $p < 0.001$; Fig. 8a). Sediment C/N ratios showed opposite, yet consistent significant trends with OM, with the lowest average C/N ratio in the

Benthic Only treatments (39.3 ± 2.57), followed by Sediment + Benthic (70.8 ± 1.17) and Sediment Only (101.4 ± 3.91; emmeans; $p < 0.001$; Fig. 8b). Across the treatments, greater percent OM and lower C/N values generally correlated with higher cumulative production across both temperature controls (Fig. 8). The difference in cumulative production between 20°C and 30°C was highest in the Benthic Only treatment (112.9 nmol CH_4 g DW^{-1}) followed by Sediment + Benthic (77.7 nmol CH_4 g DW^{-1}) and then minimal changes in production for the Sediment Only treatment (5.6 nmol CH_4 g DW^{-1}).

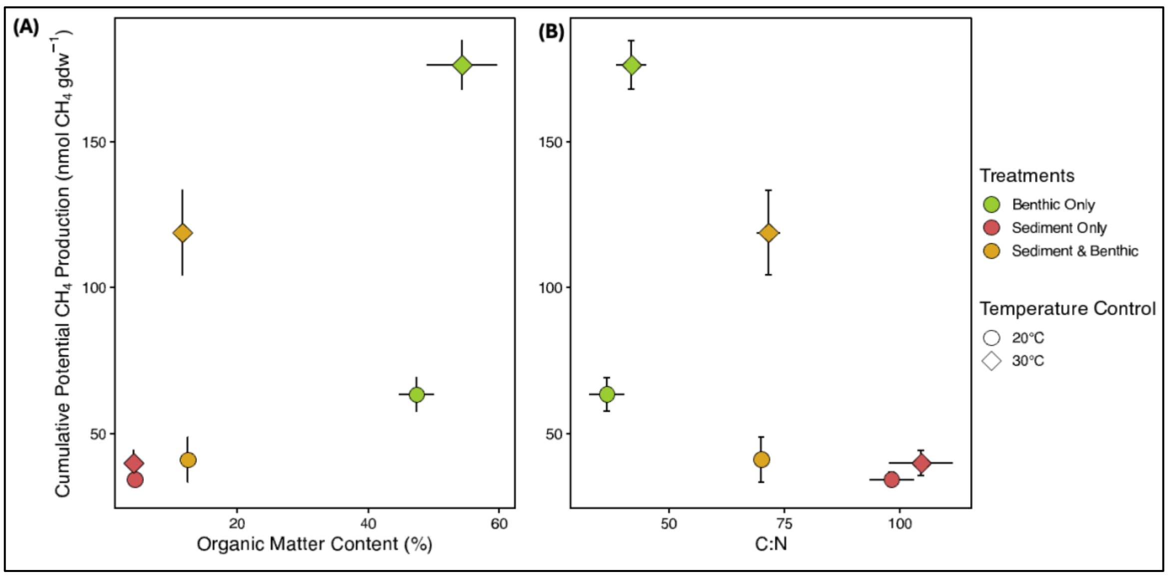

Figure 8 - Organic Matter Content and Lability have strong influence over Cumulative CH_4 Production at higher temperatures. (A) Averaged Organic Matter Content and (B) C/N Ratios compared to the cumulative CH_4 produced across chosen sample replicates split by temperature controls. Points represent the averaged values of OM content or C/N ratios and Cumulative Production from chosen replicates. Bars represent the standard error of averaged values and grey arrows represent general trends of potential CH_4 production across treatments.

2.4 Discussion

We found that littoral areas with emergent vegetation had 3.8x higher CH_4 flux than paired open water areas in our northern boreal lake (Fig. 4). The only significant relationship we found was between depth and open water ebullition (Fig. 6). We also found that potential CH_4 production was highest and most sensitive to temperature from Benthic Only treatments compared to Sediment Only and Sediment + Benthic treatments (Fig. 7). Together, our results suggest that under a shifting climate that promotes the occupation of shorelines by emergent vegetation, greater OM additions by vegetation, and warmer lake conditions, we are likely to see an increase in CH_4 emission from northern lakes. Below we explored these findings in more detail, including a comparison with the literature and an exploration of future implications of our results.

2.4.1 Flux Comparison with Literature

Fluxes from vegetated areas in our study (86.2 ± 57.3 mg C m^{-2} d^{-1}) are within the range of previously reported northern emergent plant emissions (6-167 mg C m^{-2} d^{-1}; Juutinen et al., 2004; Kankaala et al., 2005; Kyzivat et al., 2022). We found that vegetated areas have 3.8 times the flux than open water flux, which is in the mid-range of previous reports (~1-9x; Desrosiers et al., 2022; Kyzivat et al., 2022; Liu et al., 2025). Diffusive open water fluxes in our study (22.7 ± 19.5 mg C m^{-2} d^{-1}) fell within the middle of fluxes reported by past studies for other glacial and northern lakes (~9-40 mg C m^{-2} d^{-1}; Kuhn et al., 2021; Kyzivat et al., 2022; Wik et al., 2016; Fig. 9a). Ebullition was the least significant pathway in our system, with a similar contribution to overall flux (~19-48%) to what has been found previously in northern and temperate systems (18-54%; DelSontro et al., 2016; Wik et al., 2016), and across a synthesis of northern lake fluxes (11-86%; Kuhn et al., 2021). Similarly, our ebullitive flux magnitudes from open (22.0 ± 28.0 mg C m^{-2} d^{-1}) and vegetated sites (18.1 ± 28.7 mg C m^{-2} d^{-1}) fell within the lower range of reported ebullitive emissions from other northern temperate and glacial lakes (0-79 mg C m^{-2} d^{-1}; DelSontro et al., 2016; Kuhn et al., 2021).

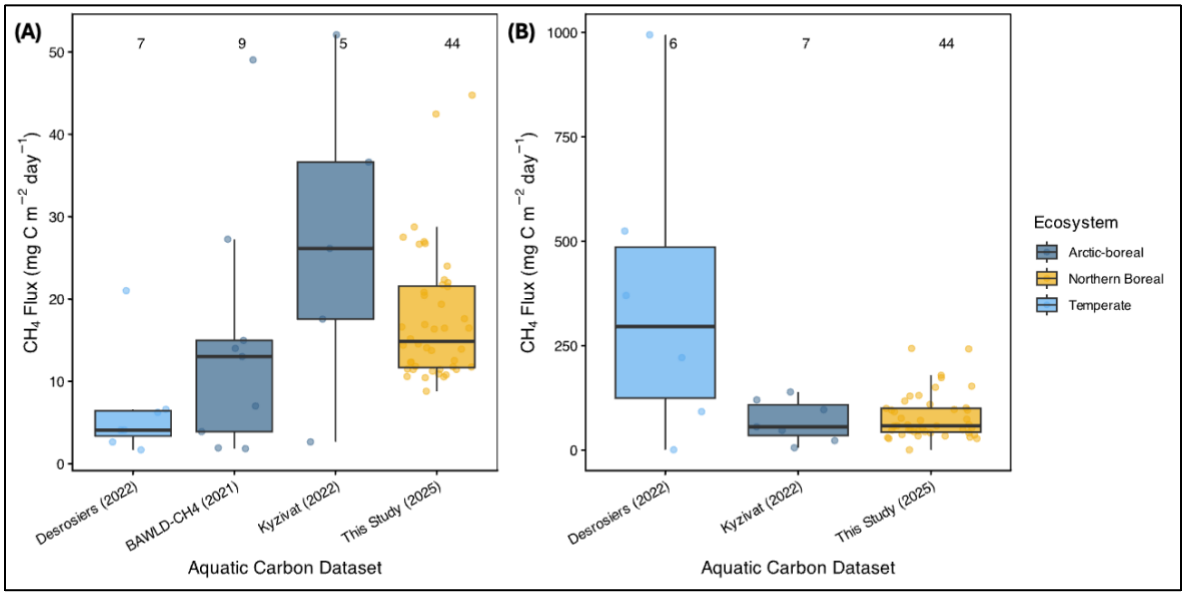

Figure 9 - Study comparison displays an underrepresentation of the influence of aquatic vegetation on CH_4 flux. Open water (A) and vegetated (B) CH_4 flux estimates from studies across Northern ecosystems. Vegetated data represents values from emergent vegetation only, and extreme outliers were removed following author's protocols. Note the different y-axis ranges across the panels. No vegetated fluxes were included in BAWLD-CH_4.

2.4.2 Emergent Vegetation Enhances Littoral CH_4 Flux

We found that plant-mediated fluxes were 3.8 times greater than adjacent open waters (Fig. 4). Our results support several previous studies that show littoral emergent vegetation enhances CH_4 emission (Bergström et al., 2007; Juutinen et al., 2003; Kyzivat et al., 2022). The increase we identified when aquatic vegetation is present is in the mid-range of past reports, which showed increases from 1 to 9 times what was reported from their open water estimates (Desrosiers et al., 2022; Kyzivat et al., 2022; Liu et al., 2025). The differences between the vegetation to open water ratios in our study and the reports from others could be due to variation in open water diffusive flux magnitudes (i.e., in the case of Kyzivat et al., 2022; Fig. 9) or changes in sediment chemistry effects and transport pathway mechanisms among different emergent vegetation species (Chanton, 2005).

Differences in plant-mediated OM availability and CH_4 production/oxidation rates could explain variation in emergent vegetation flux between different studies (Emilson et al., 2018; Grasset et al., 2019; Kankaala & Bergström, 2004). For example, Emilson et al. (2018) found that *Typha* litter increased sediment CH_4 production in northern temperate lakes by up to 400 times that of un-amended treatments, increasing the transport potential from these species. However, wetland studies have found that tussock forming species, such as *C. aquatilis*, can often have higher CH_4 fluxes than non-tussock graminoids (i.e., *Typha*) due to smaller belowground biomass which limits radial oxygen loss, leading to lower sediment oxidation and, in response, greater CH_4 production (Kao-Kniffin et al., 2010; Striker, 2023). While there's conflicting opinions on the vegetation-sediment interactions, our findings likely indicate that sediments occupied by *Carex* could be inhibiting potential CH_4 production through the oxidization of sediments by roots or by not providing sufficient substrate to an already nutrient-limited system, in turn impacting the amount of CH_4 that can be transported through the plant itself.

For example, at our site *Carex aquatilis* is the dominant species, which transport CH_4 through their aerenchyma passively, depending on concentration gradients within the plant (Chanton, 2005; Chanton & Whiting, 1996). Diffusive plant transport decreases $\delta^{13}C$-CH_4 values on the order of magnitude of -10‰ due to preferential transport of the lighter ^{12}C-CH_4 during diffusion (Chanton, 2005; Popp et al., 1999; Thompson et al., 2018). In vegetated plots, we found a difference in $\delta^{13}C$-CH_4 of -16‰ between vegetated chamber samples and source bubbles, highlighting this passive fractionation effect. However, many studies focus on species which employ active bulk transport mechanisms (i.e., *Typha* in Desrosiers et al. (2022)), wherein gas exchange is controlled by pressure gradients formed through thermal transpiration (Armstrong et al., 1996). As such, bulk transport is a non-fractionating process, leading to little difference in isotope values, and can potentially lead to higher CH_4 emissions than passive pathways due to greater gas transport particularly during peak daytime hours (Juutinen et al., 2004; Vroom et al., 2022). As many studies measure during peak daytime hours, when bulk transport tends to move gases in higher quantities, our lower fluxes are not abnormal, instead they reflect the measurement bias towards high-emission times and the possible limiting effect molecular diffusion has on plant-mediated CH_4 emission.

2.4.3 Negligible Environmental Controls on Fluxes

Often, CH_4 fluxes are strongly and positively correlated with sediment temperature (DelSontro et al., 2016; Juutinen et al., 2004; Wik et al., 2018). This well-established relationship revolves around the connection between greater microbial activity, and subsequent CH_4 production, under warmer conditions, particularly when supplied with greater, more labile, OM substrates (Duc et al., 2010; Fuchs et al., 2016; Grasset et al., 2019). Our three day averaged sediment temperatures ranged from 16 to 25 °C, which were on the high end of ranges in previously reported studies (4-24 °C; DelSontro et al., 2016; Kankaala et al., 2004; Wik et al., 2013). While within the range of past studies, we did not identify any significant relationships with temperature and fluxes, which may be due to a handful of reasons. First, we may not have adequately captured the low and high temperature ranges that occur in early summer and autumn in our lake. Second, it's possible that due to limited substrate availability or more recalcitrant OM availability methanogenesis does not respond as strongly to warmer conditions (Chanton et al., 1993; Desrosiers et al., 2022). However, we do see positive trends in the relationship between temperature and open water flux in our study, so it is possible that with more sampling this relationship will become statistically significant.

Another common environmental parameter associated with CH_4 is a negative relationship with depth, which we witness in open water areas with ebullition (Fig. 6C). This relationship is expected, because bubbling is influenced by changes in hydrostatic pressure, where greater pressure inhibits the amount of bubbling possible (Wik et al., 2013). While depth did not explain variation in plant-mediated, chamber, or vegetative ebullitive CH_4 flux significantly, we do see a negative trend in most models. Plant-mediated fluxes have been shown to follow a variable pattern with depth, where previous studies have identified non-linear (Zhao et al., 2023) and at times positive trends (Afreen et al., 2007). This variability in vegetation-depth interactions, and the other trends we've identified, suggest that a more thorough measurement of depth could reveal similar, or novel, trends to past studies particularly if our depth range is expanded.

Plant abundance metrics (i.e., biomass or density) are often noted as having strong, positive, correlations with CH_4 flux, as where there is larger biomass, there are likely larger tissues that can transport greater amounts of gas (Juutinen et al., 2003; Kankaala et al., 2005). We did not observe this relationship potentially because our biomass (27.9-207.3 g DW m^{-2}) fell within the lower end of biomass ranges reported by other studies (25-600 g DW m^{-2}; Juutinen et al., 2003; Kankaala et al., 2005; Whiting & Chanton, 1992). With smaller biomass, there is a chance that our plants had lower transport potential than what other studies have found, failing to reveal a strong, positive, relationship with biomass.

2.4.4 Potential CH_4 production increased with benthic vegetation additions

We found that potential CH_4 production treatments with benthic vegetation additions had higher OM content, lower C/N ratios, and greater potential CH_4 production rates across the 10-day experiment (Fig. 6 and 7). Benthic-only treatments had on average 60-85% greater potential production rates than the other treatments, with the most notable differences between Sediment

Only and Benthic Only treatments. Our results are consistent with other studies that have shown that labile OM availability increases sediment CH_4 production (Duc et al., 2010; Moras et al., 2024; Valentine et al., 1994). Interestingly, our C/N ratios (32.7 for Benthic Only to 112 for Sediment Only) are ~4-12 times higher than average C/N ratios reported from other northern lakes (~9-23; Duc et al., 2010; Moras et al., 2024), suggesting extreme nitrogen limitation in our system. These high C/N ratios imply that there is poor sediment substrate availability for microbial mineralization and with the addition of relatively higher OM and lower C/N benthic vegetation material, production rates increase substantially. Other studies working with more labile substrates found that autochthonous OM-additions lead to 10 times greater potential CH_4 production than sediment only treatment (Grasset et al., 2018). Furthermore, Emilson et al. (2018) found that *Typha* substrate additions in temperate lakes lead to 400 times greater production rates than sediment treatments without *Typha* additions. Given our benthic vegetation additions consisted of moss species, which typically have lower OM quality than graminoids and decompose at slower rates (Turetsky, 2003), it is understandable that vegetation additions in our study did not lead to increases in production rate as high as observed in other studies. However, it's notable for our high C/N sediments that even additions of typically more recalcitrant benthic mosses, still has the potential to substantially increase CH_4 production.

While our incubation only assessed the impact of uprooted benthic vegetation, *in situ* studies have shown that benthic vegetation can actively promote oxic CH_4 production by providing anoxic conditions within their structures allowing for methanogens to persist in typically well oxygenated settings (Hilt et al., 2022). For example, Kanaparthi et al. (2017) found that terrestrial feather mosses host methanogens and when placed under anoxic conditions with water additions they produced greater amounts of CH_4 than found in aerobic treatments. However, it is notable that many studies have also shown that submerged intact mosses promote the oxidation of CH_4 (Kox et al., 2021; Liebner et al., 2011; Sorrell et al., 2002) suggesting important mechanistic differences between active vs. decomposing submerged plant materials. Our findings indicate that during senescence there are likely high amounts of CH_4 being produced and emitted from vegetated systems, particularly when benthic vegetation is present. This adds to growing knowledge on the role of benthic vegetation in CH_4 cycling and provides support for these species to act as producers of CH_4 alongside oxidizers. More research is required to understand the role of rooted, live benthic vegetation across seasons to understand how their contributions pre- and post-senescence may alter sediment CH_4 production and consumption.

2.4.5 Benthic vegetation additions increase temperature sensitivity of methane production

We found that benthic vegetation presence increased the temperature sensitivity of CH_4 production. The Q_{10} value for the Benthic Only treatment (9.8 ± 7.2) was two times higher than the Sediment Only treatments (4.7 ± 2.8). Our observed Q_{10} values were higher than what has been reported from other CH_4 incubation studies of northern sediments (1.87 to 2.8; Duc et al., 2010; Iwata et al., 2020; Liikanen et al., 2002). However, our Q_{10} values for the benthic additions are similar to the ranges reported for *in situ* temperature sensitivity of CH_4 emission northern lakes, ponds, and wetlands (Kuhn et al., 2021; Mu et al., 2025; Rasilo et al., 2015; Turetsky et al., 2014) suggesting our vegetation additions might more closely reflect ecosystem level temperature sensitivity. With the predicted increases in lake bottom warming and consequent increases in CH_4 production (Jansen et al., 2022), our results indicate that lake shorelines are likely to become larger contributors of CH_4 due to enhanced microbial activity with OM additions. When we consider this trend alongside aquatic macrophyte expansion (Alahuhta et al., 2011; Liu et al., 2025), we are likely to see an even greater increase in CH_4 production and emission.

2.4.6 Limitations and Future Research Directions

While our study provides a detailed examination of six weeks of field-measured CH_4 emissions, future studies would benefit from using novel measurement techniques (i.e., stacked chambers; Bodmer et al., 2024) to be able to better parse apart sediment, open water, and plant-mediated fluxes in vegetated zones. We assumed diffusive flux was the same in paired open water and vegetated sites following Desrosiers et al. (2022), but this potentially masked the indirect impact of vegetation on diffusive CH_4 flux, such as changes in sediment CH_4 production potential and oxidation potential through macrophytes (Emilson et al., 2018; Heilman & Carlton, 2001). For example, surface CH_4 concentrations trended non-significantly towards being higher in vegetated vs open water areas, suggesting potential differences in diffusive emissions. Additionally, being able to note the species-specific differences in plant-mediated CH_4 flux is crucial, as it's clear that functional group is not enough to fully explain the variation in CH_4 emission within littoral areas based on the large range of CH_4 flux noted by emergent vegetation (Desrosiers et al., 2022; Kankaala et al., 2005; Kyzivat et al., 2022). Furthermore, incorporating clear chamber designs may be able to capture more dynamic gas cycling that is likely altered under light and dark conditions.

Finally, our 10-day incubation revealed important information about potential sediment CH_4 production and temperature sensitivity in the presence of benthic matter additions. Future studies could examine how potential production changes under longer timeframes (i.e. months to years), which could provide insights into how microbial communities respond to changes in carbon inputs and warmer temperatures. This can help to unravel if there is a point where potential CH_4 production is inhibited by either process, an important question as we witness northern environments undergo accelerated warming and increased carbon inputs (Blake et al., 2015; Liu et al., 2025; Parmentier et al., 2024).

2.5 Conclusion and Implications

In our study, we found that littoral areas with emergent vegetation had 3.8 times greater CH_4 emissions than adjacent open water areas and that benthic matter additions increased potential CH_4 production rates and temperature sensitivity in littoral lake sediments. These findings highlight the need to incorporate emergent and benthic vegetation contributions into field flux estimates and models, particularly when studying highly dynamic littoral areas. Global warming-induced expansion of emergent vegetation is occurring at high rates across northern, lake-dense, regions (Liu et al., 2025), yet current studies still often neglect or simplify the dynamic nature of aquatic vegetation in their estimates of CH_4 budgets. We have shown the importance of accounting for vegetation in process-based models and large synthesis studies, as their neglect leads to an underestimation of CH_4 emissions from northern lakes, particularly those that have been previously noted as being "low emitting" such as glacial and post-glacial lakes. With sustained warming, emergent vegetation is likely to maintain its expansion northward, and in response there needs to be a greater global effort towards accounting for their impacts on CH_4 cycling in aquatic systems.

Chapter 3: Conclusion

My study details the significant influence that vegetation, in both emergent and submerged forms, has on methane (CH_4) emission and production. Our field-based study focused on plant-mediated CH_4 transport from *Carex aquatilis* and highlighted the strength of emergent vegetation to emit CH_4 in large quantities, ~4 times the amount of open littoral areas. My findings support the growing knowledge on the importance of including emergent vegetation in CH_4 budget estimations. My lab-based incubations broached the question of how sediment CH_4 production may be altered by the presence of benthic mosses. I found that benthic moss additions support CH_4 production in the absence of sediment, enhance CH_4 when combined with sediments, and are highly sensitive to changes in temperature. By combining both field- and lab-based methods we can gain a mechanistic view into the role of vegetation as both a driver and transport pathway for CH_4, providing insight into the specific way that CH_4 is produced and moves through littoral areas.

3.1 Implications

My findings have implications for the understanding of CH_4 exchange in northern aquatic systems as I found that vegetation not only increases CH_4 flux directly, but also that the inputs from vegetation into sediments leads to higher potential CH_4 production that is highly sensitive to temperature. Many studies have alluded to the fact that when lake bottoms warm, we expect to see higher CH_4 emissions (Jansen et al., 2022; Kuhn et al., 2025), but the influence of vegetation is rarely mentioned. By excluding vegetation in these large-scale studies, there are key mechanisms that are missing that may lead to an underestimation of CH_4 flux from these regions that are some of the first to experience extreme impacts of global warming (Rantanen et al., 2022). As a result, we are witnessing the convergence of an expansion of aquatic vegetation into shorelines and lake bottoms warming, creating an ideal environment for promoted CH_4 production and emission.

Many incubation studies have focused on understanding how additions of organic matter, through terrestrial or macrophyte-derived litter, alter CH_4 production and consumption, and most studies have found an increase of potential CH_4 production at large magnitudes. What is less common is looking at how potential CH_4 production rates change when decomposing substrates are on their own, without sediment interactions. In my study, I observed that aquatic mosses on their own had the highest potential production rates of CH_4 and their temperature sensitivity was the highest in comparison to the sediment treatments likely due to their highest OM content and lability based on C/N comparisons. My results suggest lake CH_4 dynamics could be altered substantially with future expected changes in the expansion of emergent aquatic vegetation (Liu et al., 2025), the browning of lakes (Kritzberg et al., 2019), and the warming of boreal lakes resulting in the loss in submerged aquatic vegetation (Lind et al., 2022). The nature of my incubation also allows me to theorize the dynamics of potential CH_4 production tied during senescence at the end of the growing season (i.e., autumn) or when submerged macrophytes are out competed by their emergent counterparts. These are dynamics that haven't been heavily studied, where many studies have concentrated on simulating *in situ* conditions during growing season and understanding substate-specific potential CH_4 production (Grasset et al., 2018; Harpenslager et al., 2022). Our results on the other hand likely indicate that senescence will be a period of high CH_4 emission due to the increased additions of labile substrate that CH_4 producing archaea are eager to consume, and thus an influx of CH_4 into the system.

Current vegetation studies tend to focus on characteristic high-emitting, commonly distributed species, such as *Typha Latifolia* or the invasive *Phragmites Australis* (Bergström et al., 2007; Desrosiers et al., 2022; Kankaala et al., 2005). While these species have been shown to emit high quantities of CH_4 through their pressure and temperature driven gas exchange mechanisms, these are not the only species capable of emitting high amounts of CH_4. The *Carex* family is a common wetland species that may begin occupying more lakes that are in areas with an abundance of wetlands, such as Canada's northern boreal region, and while these species employ a gas exchange mechanism that is not as reliant on pressure or temperature differences, they can still exchange high amounts of CH_4 as I observed in my study. My results suggest that there needs to not only be attention towards the quantification and eventual management of invasive or characteristically high-emitting species, but we need to account for the variation in vegetation along lake shorelines and understand how these stands may be responding differently to environmental change.

3.2 Future Research Directions

My study was able to capture clear differences in CH_4 emission between open water and vegetated sites, but future studies can improve upon our methods and previous studies through a variety of approaches. For example, I used an approach that assumed the same diffusive flux in both open water and vegetated areas to parse apart this pathway. This assumption may not show the indirect impact of vegetation on diffusive CH_4 flux, which is likely to differ based on changes in sediment CH_4 production potential and oxidation potential of macrophytes (Emilson et al., 2018; Heilman & Carlton, 2001; Hilt et al., 2022). While not perfect, my approach allowed me to understand relative contributions of vegetation to overall flux, where future studies asking similar questions would benefit from methodologies that are able to more accurately estimate plant-mediated flux, potentially through a stacked chamber method as presented by Bodmer et al. (2024). This is important especially for studies that are trying to understand how CH_4 is being split across pathways, which can be supported using sediment, water column, and air stable C-CH_4 isotope samples. It's important for future studies to also identify species-specific differences, as it's clear that functional group is not enough to fully explain the variation in CH_4 emission within littoral areas based on the large range of CH_4 flux noted by emergent vegetation (Desrosiers et al., 2022; Kankaala et al., 2005; Kyzivat et al., 2022).

Further, while field measurements are intensive, better representation across space and time is needed to improve our understanding of dynamic plant-mediated transport. In my study, I showed the importance of taking many measurements over the span of one growing season so that we properly try to disentangle and identify what is driving vegetated flux in our study. In my case, by addressing the fine-scale differences created by a vegetated littoral areas, we may have found evidence that these environmental drivers, such as sediment temperature or depth, are more important in temporally and spatially expansive studies. Capacity permitting, future studies would benefit from established sites they can return to capture interannual changes, particularly in ecosystems and regions that are underrepresented but shown to be impacted greatly by climate induced global warming, such as northern boreal freshwater lakes (Keller, 2007; Rantanen et al., 2022; Rosentreter et al., 2021).

Finally, using a short-term incubation design, I was able to identify the general trends in CH_4 production as OM (via plant additions) and temperature changed. We used detached plant matter which means that my design gives insight into CH_4 production along this lake shore when senescence begins, but it is unable to fully understand how living mosses interact with CH_4, which is likely different to our results compared to past studies (Kox et al., 2021; Liebner et al., 2011; Struik et al., 2026). Our incubation was also relatively short, where longer incubations (i.e., >60 days) display a plateau effect on CH_4 production after 40 days (Grasset et al., 2018), potentially due to a reduction in substrate availability over time. Future studies may benefit from running a combination of shorter and longer incubations to understand the full extent of organic additions and to confirm that high production rates are relatively consistent across both earth stages and the plateau period.

3.3 Concluding Remarks

My study was able to identify that emergent vegetation drives increased CH_4 flux along vegetated lake shorelines, with increases in CH_4 flux of nearly ~4 times what is found in open water areas. Not only did emergent vegetation contribute to increases in CH_4 flux, but I also found that incubations containing benthic moss on its own and with sediments increased potential CH_4 production rates and temperature sensitivity. My findings likely indicate that the exclusion of emergent and benthic vegetation in current northern CH_4 budgets means we are likely missing large sources of CH_4. It's important for future studies to incorporate vegetation dynamics in CH_4 studies as we are currently seeing the expansion of aquatic vegetation occurring at high rates across lake-dense, regions such as Canada's northern boreal region (Liu et al., 2025). This is of particular importance for studies that are measuring from stereotypically low emitting systems, such as glacial lakes, which are projected to warm and become more hospitable to large groups of vegetation (Alahuhta et al., 2011). By putting forward a stronger, collective effort to account for the impact of vegetation on CH_4 cycling, we can create more accurate budgets and gain a clearer picture into how global warming will impact these highly sensitive northern systems.

Works Cited

Afreen, F., Zobayed, S. M. A., Armstrong, J., & Armstrong, W. (2007). Pressure gradients along whole culms and leaf sheaths, and other aspects of humidity-induced gas transport in Phragmites australis. *Journal of Experimental Botany*, *58*(7), 1651–1662.

Alahuhta, J., Heino, J., & Luoto, M. (2011). Climate change and the future distributions of aquatic macrophytes across boreal catchments. *Journal of Biogeography*, *38*(2), 383–393. https://doi.org/10.1111/j.1365-2699.2010.02412.x

Al-Haj, A. N., Chidsey, T., & Fulweiler, R. W. (2022). Two temperate seagrass meadows are negligible sources of methane and nitrous oxide. *Limnology and Oceanography*, *67*(S2), S193–S207. https://doi.org/10.1002/lno.12250

Armstrong, W. (1980). Aeration in Higher Plants. In *Advances in Botanical Research* (Vol. 7, pp. 225–332). Academic Press. https://doi.org/10.1016/S0065-2296(08)60089-0

Armstrong, W., Armstrong, J., & Beckett, P. M. (1996). Pressurised aeration in wetland macrophytes: Some theoretical aspects of humidity-induced convection and thermal transpiration. *Folia Geobotanica*, *31*(1), 25–36. https://doi.org/10.1007/BF02803991

Bastviken, D., Cole, J., Pace, M., & Tranvik, L. (2004). Methane emissions from lakes: Dependence of lake characteristics, two regional assessments, and a global estimate. *Global Biogeochemical Cycles*, *18*(4). https://doi.org/10.1029/2004GB002238

Bastviken, D., Ejlertsson, J., & Tranvik, L. (2002). Measurement of Methane Oxidation in Lakes: A Comparison of Methods. *Environmental Science & Technology*, *36*(15), 3354–3361. https://doi.org/10.1021/es010311p

Bastviken, D., Tranvik, L. J., Downing, J. A., Crill, P. M., & Enrich-Prast, A. (2011). Freshwater Methane Emissions Offset the Continental Carbon Sink. *Science*, *331*(6013), 50–50.

Bastviken, D., Treat, C. C., Pangala, S. R., Gauci, V., Enrich-Prast, A., Karlson, M., Gålfalk, M., Romano, M. B., & Sawakuchi, H. O. (2023). The importance of plants for methane emission at the ecosystem scale. *Aquatic Botany*, *184*, 103596. https://doi.org/10.1016/j.aquabot.2022.103596

Bergström, I., Mäkelä, S., Kankaala, P., & Kortelainen, P. (2007). Methane efflux from littoral vegetation stands of southern boreal lakes: An upscaled regional estimate. *Atmospheric Environment*, *41*(2), 339–351. https://doi.org/10.1016/j.atmosenv.2006.08.014

Bertolet, B. L., West, W. E., Armitage, D. W., & Jones, S. E. (2019). Organic matter supply and bacterial community composition predict methanogenesis rates in temperate lake sediments. *Limnology and Oceanography Letters*, *4*(5), 164–172. https://doi.org/10.1002/lol2.10114

Blake, L. I., Tveit, A., Øvreås, L., Head, I. M., & Gray, N. D. (2015). Response of Methanogens in Arctic Sediments to Temperature and Methanogenic Substrate Availability. *PLOS ONE*, *10*(6), e0129733. https://doi.org/10.1371/journal.pone.0129733

Bodmer, P., Vroom, R. J. E., Stepina, T., del Giorgio, P. A., & Kosten, S. (2024). Methane dynamics in vegetated habitats in inland waters: Quantification, regulation, and global significance. *Frontiers in Water*, *5*. https://doi.org/10.3389/frwa.2023.1332968

Bodmer, P., Vroom, R., Stepina, T., Giorgio, P. A. del, & Kosten, S. (2021). *Methane Fluxes of Vegetated Areas in Natural Freshwater Ecosystems: Assessments and Global Significance*. https://eartharxiv.org/repository/view/2904/

Brooks, M., Bolker, B., Kristensen, K., Maechler, M., Magnusson, A., McGillycuddy, M., Skaug, H., Nielsen, A., Berg, C., Bentham, K. van, Sadat, N., Lüdecke, D., Lenth, R., O'Brien, J., Geyer, C. J., Jagan, M., Wiernik, B., Stouffer, D. B., Agronah, M., …

Williams, C. (2026). *glmmTMB: Generalized Linear Mixed Models using Template Model Builder* (Version 1.1.14) [Computer software]. https://cran.r-project.org/web/packages/glmmTMB/index.html

Carmichael, M. J., Bernhardt, E. S., Bräuer, S. L., & Smith, W. K. (2014). The role of vegetation in methane flux to the atmosphere: Should vegetation be included as a distinct category in the global methane budget? *Biogeochemistry*, *119*(1), 1–24. https://doi.org/10.1007/s10533-014-9974-1

Chanton, J. P. (2005). The effect of gas transport on the isotope signature of methane in wetlands. *Organic Geochemistry, Stable Isotope Applications in Methane Cycle Studies*, *36*(5), 753–768. https://doi.org/10.1016/j.orggeochem.2004.10.007

Chanton, J. P., & Whiting, G. J. (1996). Methane stable isotopic distributions as indicators of gas transport mechanisms in emergent aquatic plants. *Aquatic Botany, Ventilation in Macrophytes*, *54*(2), 227–236. https://doi.org/10.1016/0304-3770(96)01047-9

Chanton, J. P., Whiting, G. J., Happell, J. D., & Gerard, G. (1993). Contrasting rates and diurnal patterns of methane emission from emergent aquatic macrophytes. *Aquatic Botany*, *46*(2), 111–128. https://doi.org/10.1016/0304-3770(93)90040-4

Cole, J. J., Prairie, Y. T., Caraco, N. F., McDowell, W. H., Tranvik, L. J., Striegl, R. G., Duarte, C. M., Kortelainen, P., Downing, J. A., Middelburg, J. J., & Melack, J. (2007). Plumbing the Global Carbon Cycle: Integrating Inland Waters into the Terrestrial Carbon Budget. *Ecosystems*, *10*(1), 172–185. https://doi.org/10.1007/s10021-006-9013-8

Deemer, B. R., & Holgerson, M. A. (2021). Drivers of Methane Flux Differ Between Lakes and Reservoirs, Complicating Global Upscaling Efforts. *Journal of Geophysical Research: Biogeosciences*, *126*(4), e2019JG005600. https://doi.org/10.1029/2019JG005600

DelSontro, T., Boutet, L., St-Pierre, A., del Giorgio, P. A., & Prairie, Y. T. (2016). Methane ebullition and diffusion from northern ponds and lakes regulated by the interaction between temperature and system productivity. *Limnology and Oceanography*, *61*(S1), S62–S77. https://doi.org/10.1002%2Flno.10335

Desrosiers, K., DelSontro, T., & del Giorgio, P. A. (2022). Disproportionate Contribution of Vegetated Habitats to the CH4 and CO2 Budgets of a Boreal Lake. *Ecosystems*, *25*(7), 1522–1541. https://doi.org/10.1007/s10021-021-00730-9

Drake, T. W., Raymond, P. A., & Spencer, R. G. M. (2018). Terrestrial carbon inputs to inland waters: A current synthesis of estimates and uncertainty. *Limnology and Oceanography Letters*, *3*(3), 132–142. https://doi.org/10.1002/lol2.10055

Duc, N. T., Crill, P., & Bastviken, D. (2010). Implications of temperature and sediment characteristics on methane formation and oxidation in lake sediments. *Biogeochemistry*, *100*(1), 185–196. https://doi.org/10.1007/s10533-010-9415-8

Emilson, E. J. S., Carson, M. A., Yakimovich, K. M., Osterholz, H., Dittmar, T., Gunn, J. M., Mykytczuk, N. C. S., Basiliko, N., & Tanentzap, A. J. (2018). Climate-driven shifts in sediment chemistry enhance methane production in northern lakes. *Nature Communications*, *9*(1), 1801. https://doi.org/10.1038/s41467-018-04236-2

Fuchs, A., Lyautey, E., Montuelle, B., & Casper, P. (2016). Effects of increasing temperatures on methane concentrations and methanogenesis during experimental incubation of sediments from oligotrophic and mesotrophic lakes. *Journal of Geophysical Research: Biogeosciences*, *121*(5), 1394–1406. https://doi.org/10.1002/2016JG003328

Grasset, C., Abril, G., Mendonça, R., Roland, F., & Sobek, S. (2019). The transformation of macrophyte-derived organic matter to methane relates to plant water and nutrient contents. *Limnology and Oceanography*, *64*(4), 1737–1749. https://doi.org/10.1002/lno.11148

Grasset, C., Mendonça, R., Saucedo, G. V., Bastviken, D., Roland, F., & Sobek, S. (2018). Large but variable methane production in anoxic freshwater sediment upon addition of allochthonous and autochthonous organic matter. *Limnology and Oceanography*, *63*(4), 1488–1501.

Harpenslager, S. F., Thiemer, K., Levertz, C., Misteli, B., Sebola, K. M., Schneider, S. C., Hilt, S., & Köhler, J. (2022). Short-term effects of macrophyte removal on emission of CO2 and CH4 in shallow lakes. *Aquatic Botany*, *182*, 103555. https://doi.org/10.1016/j.aquabot.2022.103555

Heilman, M. A., & Carlton, R. G. (2001). Methane Oxidation Associated with Submersed Vascular Macrophytes and Its Impact on Plant Diffusive Methane Flux. *Biogeochemistry*, *52*(2), 207–224.

Heiri, O., Lotter, A. F., & Lemcke, G. (2001). Loss on ignition as a method for estimating organic and carbonate content in sediments: Reproducibility and comparability of results. *Journal of Paleolimnology*, *25*(1), 101–110. https://doi.org/10.1023/A:1008119611481

Henneberg, A., Brix, H., & Sorrell, B. K. (2015). The interactive effect of Juncus effusus and water table position on mesocosm methanogenesis and methane emissions. *Plant and Soil*, *400*(1), 45–54. https://doi.org/10.1007/s11104-015-2707-y

Hershey, A. E., Northington, R. M., & Whalen, S. C. (2014). Substrate limitation of sediment methane flux, methane oxidation and use of stable isotopes for assessing methanogenesis pathways in a small arctic lake. *Biogeochemistry*, *117*(2–3), 325–336. https://doi.org/10.1007/s10533-013-9864-y

Hilt, S., Grossart, H.-P., McGinnis, D. F., & Keppler, F. (2022). Potential role of submerged macrophytes for oxic methane production in aquatic ecosystems. *Limnology and Oceanography*, *67*(S2), S76–S88. https://doi.org/10.1002/lno.12095

Hudspeth, Z. W., Morningstar, J. L., Mendlovitz, H. P., Baily, J. A., Lloyd, K. G., & Martens, C. S. (2024). In situ aerobic methane oxidation rates in a stratified lake. *Limnology and Oceanography*, *n/a*(n/a). https://doi.org/10.1002/lno.12583

Iwata, H., Nakazawa, K., Sato, H., Itoh, M., Miyabara, Y., Hirata, R., Takahashi, Y., Tokida, T., & Endo, R. (2020). Temporal and spatial variations in methane emissions from the littoral zones of a shallow mid-latitude lake with steady methane bubble emission areas. *Agricultural and Forest Meteorology*, *295*, 108184. https://doi.org/10.1016/j.agrformet.2020.108184

Jansen, J., Woolway, R. I., Kraemer, B. M., Albergel, C., Bastviken, D., Weyhenmeyer, G. A., Marcé, R., Sharma, S., Sobek, S., Tranvik, L. J., Perroud, M., Golub, M., Moore, T. N., Råman Vinnå, L., La Fuente, S., Grant, L., Pierson, D. C., Thiery, W., & Jennings, E. (2022). Global increase in methane production under future warming of lake bottom waters. *Global Change Biology*, *28*(18), 5427–5440. https://doi.org/10.1111/gcb.16298

Juutinen, S., Alm, J., Larmola, T., Huttunen, J. T., Morero, M., Martikainen, P. J., & Silvola, J. (2003). Major implication of the littoral zones for methane release from boreal lakes. *Global Biogeochemical Cycles*, *17*(4). https://doi.org/10.1029/2003GB002105

Juutinen, S., Alm, J., Larmola, T., Saarnio, S., Martikainen, P. J., & Silvola, J. (2004). Stand-specific diurnal dynamics of CH_4 fluxes in boreal lakes: Patterns and controls. *Journal of Geophysical Research: Atmospheres*, *109*(D19), 2004JD004782. https://doi.org/10.1029/2004JD004782

Kanaparthi, D., Reim, A., Martinson, G. O., Pommerenke, B., & Conrad, R. (2017). Methane emission from feather moss stands. *Global Change Biology*, *23*, 4884–4895. https://doi.org/10.1111/gcb.13764

Kankaala, P., & Bergström, I. (2004). Emission and oxidation of methane in Equisetum fluviatile stands growing on organic sediment and sand bottoms. *Biogeochemistry*, *67*(1), 21–37. https://doi.org/10.1023/B:BIOG.0000015277.17288.7a

Kankaala, P., Käki, T., Mäkelä, S., Ojala, A., Pajunen, H., & Arvola, L. (2005). Methane efflux in relation to plant biomass and sediment characteristics in stands of three common emergent macrophytes in boreal mesoeutrophic lakes. *Global Change Biology*, *11*(1), 145–153. https://doi.org/10.1111/j.1365-2486.2004.00888.x

Kankaala, P., Ojala, A., & Käki, T. (2004). Temporal and spatial variation in methane emissions from a flooded transgression shore of a boreal lake. *Biogeochemistry*, *68*(3), 297–311. https://doi.org/10.1023/B:BIOG.0000031030.77498.1f

Kao-Kniffin, J., Freyre, D. S., & Balser, T. C. (2010). Methane dynamics across wetland plant species. *Aquatic Botany*, *93*(2), 107–113. https://doi.org/10.1016/j.aquabot.2010.03.009

Karvinen, A., Lehtinen, L., & Kankaala, P. (2015). Variable Effects of Iron (Fe (III)) Additions on Potential Methane Production in Boreal Lake Littoral Sediments. *Wetlands*, *35*(1), 137–146. https://doi.org/10.1007/s13157-014-0602-6

Keller, W. (2007). Implications of climate warming for Boreal Shield lakes: A review and synthesis. *Environmental Reviews*, *15*(NA), 99–112. https://doi.org/10.1139/A07-002

Knopp, J. A., Levenstein, B., Watson, A., Ivanova, I., & Lento, J. (2022). Systematic review of documented Indigenous Knowledge of freshwater biodiversity in the circumpolar Arctic. *Freshwater Biology*, *67*(1), 194–209. https://doi.org/10.1111/fwb.13570

Kox, M. A. R., Smolders, A. J. P., Speth, D. R., Lamers, L. P. M., Op den Camp, H. J. M., Jetten, M. S. M., & van Kessel, M. A. H. J. (2021). A Novel Laboratory-Scale Mesocosm Setup to Study Methane Emission Mitigation by Sphagnum Mosses and Associated Methanotrophs. *Frontiers in Microbiology*, *12*. https://doi.org/10.3389/fmicb.2021.651103

Kritzberg, E. S., Hasselquist, E. M., Škerlep, M., Löfgren, S., Olsson, O., Stadmark, J., Valinia, S., Hansson, L.-A., & Laudon, H. (2019). Browning of freshwaters: Consequences to ecosystem services, underlying drivers, and potential mitigation measures. *Ambio*, *49*(2), 375–390. https://doi.org/10.1007/s13280-019-01227-5

Kuhn, M. A., Schmidt, M., Heffernan, L., Stührenberg, J., Knorr, K.-H., Estop-Aragonés, C., Broder, T., Gonzalez Moguel, R., Douglas, P. M. J., & Olefeldt, D. (2023). High ebullitive, millennial-aged greenhouse gas emissions from thermokarst expansion of peatland lakes in boreal western Canada. *Limnology and Oceanography*, *68*(2), 498–513. https://doi.org/10.1002/lno.12288

Kuhn, M. A., Varner, R. K., Bastviken, D., Crill, P., MacIntyre, S., Turetsky, M., Walter Anthony, K., McGuire, A. D., & Olefeldt, D. (2021). BAWLD-CH_4: A comprehensive dataset of methane fluxes from boreal and arctic ecosystems. *Earth System Science Data*, *13*(11), 5151–5189. https://doi.org/10.5194/essd-13-5151-2021

Kuhn, M., Olefeldt, D., Arndt, K. A., Bastviken, D., Bruhwiler, L., Crill, P., DelSontro, T., Fluet-Chouinard, E., Grosse, G., Hovemyr, M., Hugelius, G., MacIntyre, S., Malhotra, A., McGuire, A. D., Oh, Y., Poulter, B., Treat, C. C., Turetsky, M. R., Varner, R. K., … Zhang, Z. (2025). Current and future methane emissions from boreal-Arctic wetlands and lakes. *Nature Climate Change*, *15*(9), 986–991. https://doi.org/10.1038/s41558-025-02413-y

Kyzivat, E. D., Smith, L. C., Garcia-Tigreros, F., Huang, C., Wang, C., Langhorst, T., Fayne, J. V., Harlan, M. E., Ishitsuka, Y., Feng, D., Dolan, W., Pitcher, L. H., Wickland, K. P., Dornblaser, M. M., Striegl, R. G., Pavelsky, T. M., Butman, D. E., & Gleason, C. J. (2022). The Importance of Lake Emergent Aquatic Vegetation for Estimating Arctic-Boreal Methane Emissions. *Journal of Geophysical Research: Biogeosciences*, *127*(6), e2021JG006635. https://doi.org/10.1029/2021JG006635

Langenegger, T., Vachon, D., Donis, D., & McGinnis, D. F. (2022). Methane oxidation dynamics in a stratified lake: Insights revealed from a mass balance and carbon stable isotopes. *Limnology and Oceanography*, *67*(10), 2157–2173. https://doi.org/10.1002/lno.12195

Lenth, R. V., Piaskowski, J., Banfai, B., Bolker, B., Buerkner, P., Giné-Vázquez, I., Hervé, M., Jung, M., Love, J., Miguez, F., Riebl, H., & Singmann, H. (2026). *emmeans: Estimated Marginal Means, aka Least-Squares Means* (Version 2.0.3) [Computer software]. https://cran.r-project.org/web/packages/emmeans/index.html

Li, L., & Xue, B. (2023). Spatial-temporal variability of methane fluxes in lakes varying in latitude, area, and depth. *Heliyon*, *9*(8), e18411. https://doi.org/10.1016/j.heliyon.2023.e18411

Liebner, S., Zeyer, J., Wagner, D., Schubert, C., Pfeiffer, E.-M., & Knoblauch, C. (2011). Methane oxidation associated with submerged brown mosses reduces methane emissions from Siberian polygonal tundra. *Journal of Ecology*, *99*(4), 914–922.

Liikanen, A., Murtoniemi, T., Tanskanen, H., Väisänen, T., & Martikainen, P. J. (2002). Effects of temperature and oxygenavailability on greenhouse gas and nutrient dynamics in sediment of a eutrophic mid-boreal lake. *Biogeochemistry*, *59*(3), 269–286. https://doi.org/10.1023/A:1016015526712

Lind, L., Eckstein, R. L., & Relyea, R. A. (2022). Direct and indirect effects of climate change on distribution and community composition of macrophytes in lentic systems. *Biological Reviews*, *97*(4), 1677–1690. https://doi.org/10.1111/brv.12858

Liu, J., Huang, H., Hou, X., Feng, L., Pi, X., Kyzivat, E. D., Zhang, Y., Woodman, S. G., Tang, L., Cheng, X., & Gong, P. (2025). Expansion of aquatic vegetation in northern lakes amplified methane emissions. *Nature Geoscience*, 1–8. https://doi.org/10.1038/s41561-025-01667-7

Lobato-de Magalhães, T., Murphy, K., Efremov, A., Davidson, T. A., Molina-Navarro, E., Wood, K. A., Tapia-Grimaldo, J., Hofstra, D., Fu, H., & Ortegón-Aznar, I. (2023). How on Earth did that get there? Natural and human vectors of aquatic macrophyte global distribution. *Hydrobiologia*, *850*(7), 1515–1542. https://doi.org/10.1007/s10750-022-05107-0

Mjelde, M., Thrane, J.-E., & Demars, B. O. L. (2023). High aquatic macrophyte diversity in Norwegian lakes north of the Arctic Circle. *Freshwater Biology*, *68*(3), 509–522. https://doi.org/10.1111/fwb.14043

Moras, S., Zellmer, U. R., Hiltunen, E., Grasset, C., & Sobek, S. (2024). Predicting Methane Formation Rates of Freshwater Sediments in Different Biogeographic Regions. *Journal of Geophysical Research: Biogeosciences*, *129*(1), e2023JG007463. https://doi.org/10.1029/2023JG007463

Mu, M., Mu, C., Liu, H., Lei, P., Ge, Y., Zhou, Z., Peng, X., & Ma, T. (2025). Thermokarst lake drainage halves the temperature sensitivity of CH4 release on the Qinghai-Tibet Plateau. *Nature Communications*, *16*(1), 1992. https://doi.org/10.1038/s41467-025-57356-x

Ojala, A., Käki, T., & Huttunen, J. (2000). Methane emissions from a boreal meso-eutrophic lake: An assessment of seasonal and diel variation among emergent vegetation. *SIL Proceedings, 1922-2010*, *27*(3), 1420–1423. https://doi.org/10.1080/03680770.1998.11901470

Olefeldt, D., Hovemyr, M., Kuhn, M. A., Bastviken, D., Bohn, T. J., Connolly, J., Crill, P., Euskirchen, E. S., Finkelstein, S. A., Genet, H., Grosse, G., Harris, L. I., Heffernan, L., Helbig, M., Hugelius, G., Hutchins, R., Juutinen, S., Lara, M. J., Malhotra, A., … Watts, J. D. (2021). The Boreal–Arctic Wetland and Lake Dataset (BAWLD). *Earth System Science Data*, *13*(11), 5127–5149. https://doi.org/10.5194/essd-13-5127-2021

Parmentier, F.-J. W., Thornton, B. F., Silyakova, A., & Christensen, T. R. (2024). Vulnerability of Arctic-Boreal methane emissions to climate change. *Frontiers in Environmental Science*, *12*. https://doi.org/10.3389/fenvs.2024.1460155

Pi, X., Luo, Q., Feng, L., Xu, Y., Tang, J., Liang, X., Ma, E., Cheng, R., Fensholt, R., Brandt, M., Cai, X., Gibson, L., Liu, J., Zheng, C., Li, W., & Bryan, B. A. (2022). Mapping global lake dynamics reveals the emerging roles of small lakes. *Nature Communications*, *13*(1), 5777. https://doi.org/10.1038/s41467-022-33239-3

Popp, T. J., Chanton, J. P., Whiting, G. J., & Grant, N. (1999). Methane stable isotope distribution at a Carex dominated fen in north central Alberta. *Global Biogeochemical Cycles*, *13*(4), 1063–1077. https://doi.org/10.1029/1999GB900060

R Core Team. (2025). *R: A Language and Environment for Statistical Computing* [R]. R Foundation for Statistical Computing. https://www.R-project.org/

Rantanen, M., Karpechko, A. Y., Lipponen, A., Nordling, K., Hyvärinen, O., Ruosteenoja, K., Vihma, T., & Laaksonen, A. (2022). The Arctic has warmed nearly four times faster than the globe since 1979. *Communications Earth & Environment*, *3*(1), 1–10. https://doi.org/10.1038/s43247-022-00498-3

Rasilo, T., Ojala, A., Huotari, J., Starr, M., & Pumpanen, J. (2015). Concentrations and quality of DOC along the terrestrial–aquatic continuum in a boreal forested catchment. *Freshwater Science*, *34*(2), 440–455. https://doi.org/10.1086/680682

Rosentreter, J. A., Borges, A. V., Deemer, B. R., Holgerson, M. A., Liu, S., Song, C., Melack, J., Raymond, P. A., Duarte, C. M., Allen, G. H., Olefeldt, D., Poulter, B., Battin, T. I., & Eyre, B. D. (2021). Half of global methane emissions come from highly variable aquatic ecosystem sources. *Nature Geoscience*, *14*(4), 225–230. https://doi.org/10.1038/s41561-021-00715-2

Saarela, T., Rissanen, A. J., Ojala, A., Pumpanen, J., Aalto, S. L., Tiirola, M., Vesala, T., & Jäntti, H. (2019). CH4 oxidation in a boreal lake during the development of hypolimnetic hypoxia. *Aquatic Sciences*, *82*(2), 19. https://doi.org/10.1007/s00027-019-0690-8

Saunois, M., Martinez, A., Poulter, B., Zhang, Z., Raymond, P. A., Regnier, P., Canadell, J. G., Jackson, R. B., Patra, P. K., Bousquet, P., Ciais, P., Dlugokencky, E. J., Lan, X., Allen, G. H., Bastviken, D., Beerling, D. J., Belikov, D. A., Blake, D. R., Castaldi, S., … Zhuang, Q. (2025). Global Methane Budget 2000–2020. *Earth System Science Data*, *17*(5), 1873–1958. https://doi.org/10.5194/essd-17-1873-2025

Sorrell, B. K., Downes, M. T., & Stanger, C. L. (2002). Methanotrophic bacteria and their activity on submerged aquatic macrophytes. *Aquatic Botany*, *72*(2), 107–119. https://doi.org/10.1016/S0304-3770(01)00215-7

Striker, G. G. (2024). Oxygen Transport and Plant Ventilation. In J.-I. Sakagami & M. Nakazono (Eds.), *Responses of Plants to Soil Flooding* (pp. 139–156). Springer Nature. https://doi.org/10.1007/978-981-99-9112-9_9

Struik, Q., Groenbos, R. E., Paranaíba, J. R., Bakker, M. M., Meulepas, B. M. J. W., Lamers, L. P. M., Veraart, A. J., & Kosten, S. (2026). Macrophyte-associated methane oxidation as a key process diminishing methane emissions from agricultural drainage ditches. *Aquatic Botany*, *205*, 104007. https://doi.org/10.1016/j.aquabot.2026.104007

Theus, M. E., Ray, N. E., Bansal, S., & Holgerson, M. A. (2023). Submersed Macrophyte Density Regulates Aquatic Greenhouse Gas Emissions. *Journal of Geophysical Research: Biogeosciences*, *128*(10), e2023JG007758. https://doi.org/10.1029/2023JG007758

Thompson, H. A., White, J. R., & Pratt, L. M. (2018). Spatial variation in stable isotopic composition of organic matter of macrophytes and sediments from a small Arctic lake in west Greenland. *Arctic, Antarctic, and Alpine Research*, *50*(1), S100017. https://doi.org/10.1080/15230430.2017.1420282

Turetsky, M. R. (2003). The Role of Bryophytes in Carbon and Nitrogen Cycling. *The Bryologist*, *106*(3), 395–409. https://doi.org/10.1639/05

Turetsky, M. R., Kotowska, A., Bubier, J., Dise, N. B., Crill, P., Hornibrook, E. R. C., Minkkinen, K., Moore, T. R., Myers-Smith, I. H., Nykänen, H., Olefeldt, D., Rinne, J., Saarnio, S., Shurpali, N., Tuittila, E.-S., Waddington, J. M., White, J. R., Wickland, K.

P., & Wilmking, M. (2014). A synthesis of methane emissions from 71 northern, temperate, and subtropical wetlands. *Global Change Biology*, *20*(7), 2183–2197. https://doi.org/10.1111/gcb.12580

Valentine, D. W., Holland, E. A., & Schimel, D. S. (1994). Ecosystem and physiological controls over methane production in northern wetlands. *Journal of Geophysical Research: Atmospheres*, *99*(D1), 1563–1571. https://doi.org/10.1029/93JD00391

Villa, J. A., Ju, Y., Stephen, T., Rey-Sanchez, C., Wrighton, K. C., & Bohrer, G. (2020). Plant-mediated methane transport in emergent and floating-leaved species of a temperate freshwater mineral-soil wetland. *Limnology and Oceanography*, *65*(7), 1635–1650. https://doi.org/10.1002/lno.11467

Virkkala, A.-M., Wargowsky, I., Vogt, J., Kuhn, M. A., Madaan, S., O'Keefe, R., Windholz, T., Arndt, K. A., Rogers, B. M., Watts, J. D., Kent, K., Göckede, M., Olefeldt, D., Rocher-Ros, G., Schuur, E. A. G., Bastviken, D., Aalstad, K., Aho, K., Ala-Könni, J., … Natali, S. M. (2025). ABCFlux v2: Arctic–boreal CO_2 and CH_4 monthly flux observations and ancillary information across terrestrial and freshwater ecosystems. *Earth System Science Data Discussions*, 1–86. https://doi.org/10.5194/essd-2025-585

Vroom, R. J. E., van den Berg, M., Pangala, S. R., van der Scheer, O. E., & Sorrell, B. K. (2022). Physiological processes affecting methane transport by wetland vegetation – A review. *Aquatic Botany*, *182*, 103547. https://doi.org/10.1016/j.aquabot.2022.103547

Whiting, G. J., & Chanton, J. P. (1992). Plant-dependent CH4 emission in a subarctic Canadian fen. *Global Biogeochemical Cycles*, *6*(3), 225–231. https://doi.org/10.1029/92GB00710

Wik, M., Crill, P. M., Varner, R. K., & Bastviken, D. (2013). Multiyear measurements of ebullitive methane flux from three subarctic lakes. *Journal of Geophysical Research: Biogeosciences*, *118*(3), 1307–1321. https://doi.org/10.1002/jgrg.20103

Wik, M., Johnson, J. E., Crill, P. M., DeStasio, J. P., Erickson, L., Halloran, M. J., Fahnestock, M. F., Crawford, M. K., Phillips, S. C., & Varner, R. K. (2018). Sediment Characteristics and Methane Ebullition in Three Subarctic Lakes. *Journal of Geophysical Research: Biogeosciences*, *123*(8), 2399–2411. https://doi.org/10.1029/2017JG004298

Wik, M., Varner, R. K., Anthony, K. W., MacIntyre, S., & Bastviken, D. (2016). Climate-sensitive northern lakes and ponds are critical components of methane release. *Nature Geoscience*, *9*(2), 99–105. https://doi.org/10.1038/ngeo2578

Xu, L. (2020). *Measuring gas concentrations of small samples with Trace Gas Analyzers* [Application Note]. LICOR. https://www.licor.com/support/Trace-Gas-Analyzers/topics/measuring-small-gas-samples.html#2Closedloopmethod

Zhang, M., Xiao, Q., Zhang, Z., Gao, Y., Zhao, J., Pu, Y., Wang, W., Xiao, W., Liu, S., & Lee, X. (2019). Methane flux dynamics in a submerged aquatic vegetation zones in a subtropical lake. *Science of The Total Environment*, *672*, 400–409. https://doi.org/10.1016/j.scitotenv.2019.03.466

Zhao, M., Li, P., Song, W., Chu, X., Eller, F., Wang, X., Liu, J., Xiao, L., Wei, S., Li, X., & Han, G. (2023). Inundation depth stimulates plant-mediated CH4 emissions by increasing ecosystem carbon uptake and plant height in an estuarine wetland. *Functional Ecology*, *37*(3), 536–550. https://doi.org/10.1111/1365-2435.14258

Zuur, A. F., Ieno, E. N., & Elphick, C. S. (2010). *A protocol for data exploration to avoid common statistical problems*. https://besjournals.onlinelibrary.wiley.com/doi/10.1111/j.2041-210X.2009.00001.x

Appendices

Appendix A: Supporting Information for Chapter 2

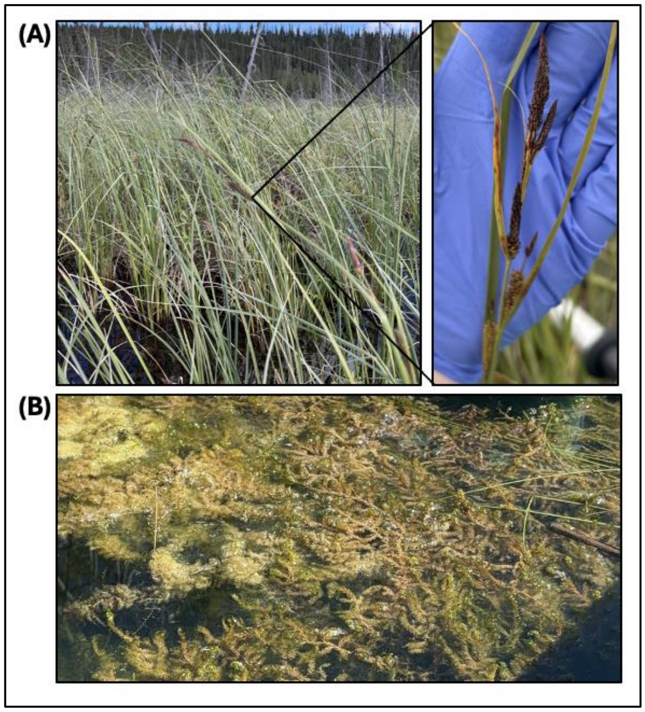

Figure S1 - Study species: (A) *Carex aquatilis* and (B) *Scorpidum* moss. Species identification was conducted using iNaturalist and by a bryologist expert at the Beaty Biodiversity Museum, University of British Columbia, Canada respectively

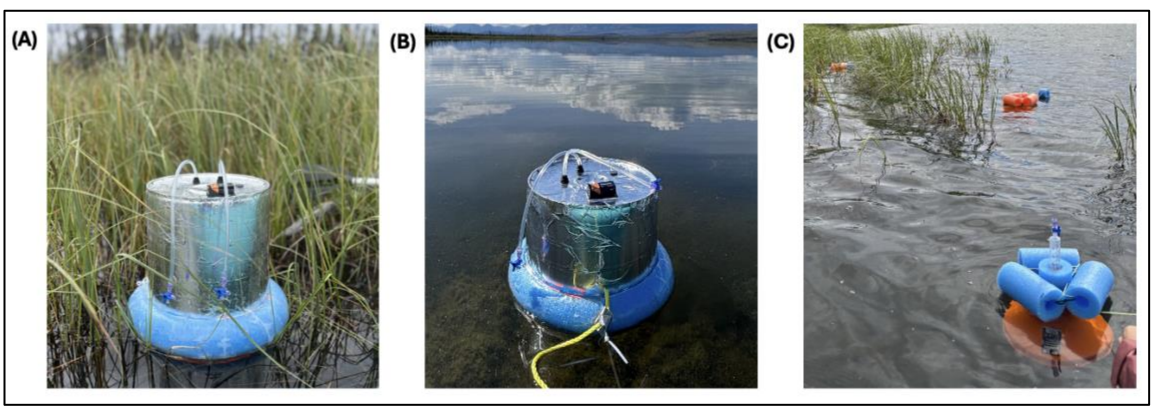

Figure S2 - Design of floating chambers (A/B) and Bubble Traps (C). Two variants of floating chambers with a taller chamber (A) created to account for the additional height needed to cover *Carex aquatilis* with little breakage and smaller chambers for open water measurements (B). Bubble traps had one design, which consisted of inverted, 25 cm diameter, funnels with a syringe and flotation device attachment.

Table S1 - Detailed Isotope Sample information. Only samples that returned sufficient values are shown below, with sample date and site provided. C represents chamber-based samples and E ebullitive samples.

Date	1AO	1AV	1BO	1BV	1CO	1CV	2AO	2AV	2BO	2BV	2CO	2CV	3AO	3AV	3BO	3BV	3CO	3CV
07/14/25			C/E	C		E			E	E								
07/16/25													E	E			C	C
07/31/25	E										E	E			E	E		
08/02/25	C	C					C	C	C	C	C	C	C	C	C	C	C	C
08/14/25	E	E							E	E							E	E
08/17/25									C	C								
08/18/25													C	C				

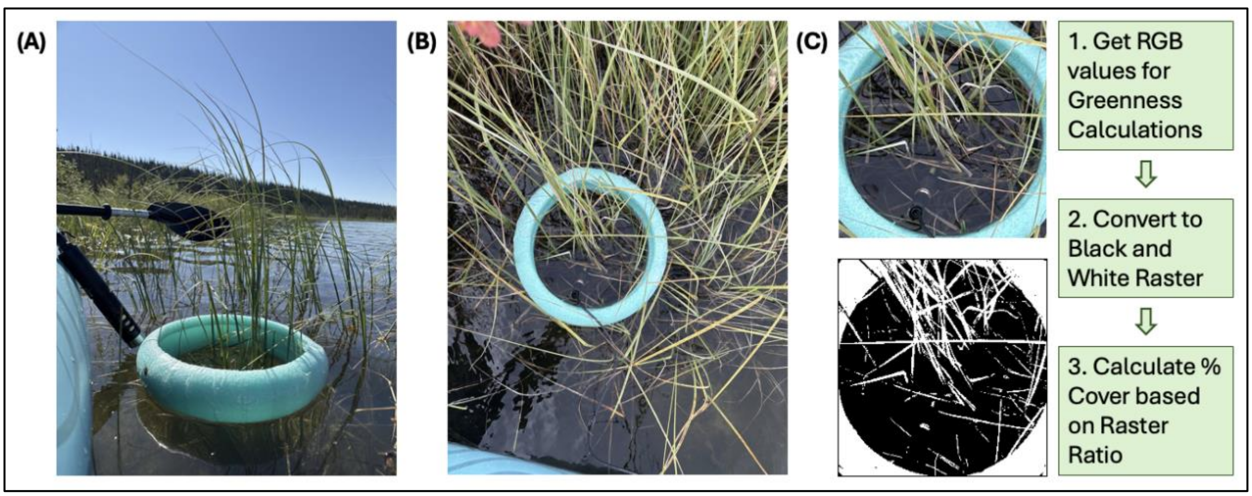

Figure S3 - Sample photos and workflow for the calculation of vegetation abundance measures and greenness indices. Side (A) and Top (B) view photos were used to understand shoot abundance within chamber areas (circle). To calculate greenness and surface area coverage, top-view images were processed in ImageJ with a general workflow noted (C).

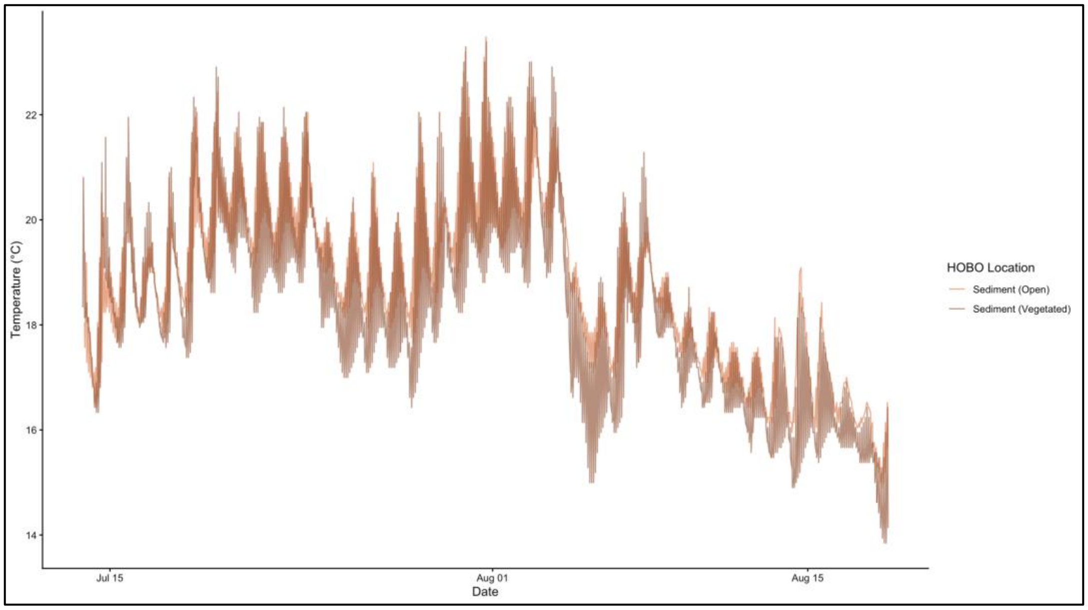

Figure S4 - 2-hourly sediment temperatures across the field season split by open and vegetated placements. Orange represents open sediment measurements, with darker brown representing vegetated measurements. Measurements captured using Onset HOBO pendants (UA-001-64; UA-002-64).

Table S2 - Fixed effect results from Generalized Linear Mixed Models. Full dataset was split into vegetated and open data to identify statistically significant relationships. We used a random effect of "site" to account for repeated measures. Additional Non-Significant Models included TDS, pH, surface water temperature, and pressure.

Emission	Parameter	n	β	SE	z-value	p-value
Diffusive Only (Open Sites)	Depth	37	0.003	0.003	0.992	0.321
	Bottom Water Temperature (3-Day Average)	60	0.022	0.021	1.016	0.309
	Sediment Temperature (3-Day Average)	60	0.025	0.027	0.964	0.335
	Dissolved Oxygen	60	0.007	0.004	1.88	0.06
	Hourly Wind Speed	60	-0.01	0.026	-0.553	0.580
	Sample Date	60	-0.001	0.003	-0.191	0.849
	Depth	39	0.0003	0.002	0.171	0.864

Diffusive + Plant-Mediated (Vegetated Sites)	Bottom Water Temperature (3-Day Average)	61	-0.009	0.021	-0.414	0.679
	Sediment Temperature (3-Day Average)	61	-0.012	0.023	-0.499	0.618
	Dissolved Oxygen	61	-0.001	0.004	-0.275	0.783
	Hourly Wind Speed	61	0.001	0.025	0.044	0.965
	Biomass	61	-0.001	0.001	-0.800	0.424
	Abovewater Volume	61	-0.001	0.001	-1.222	0.222
	Greenness Chromatic Coordinate	61	0.083	2.634	0.032	0.975
	Sample Date	61	0.003	0.003	1.026	0.305
Plant-Mediated Only (Vegetated Sites)	Depth	39	$-2x10^{-4}$	0.002	-0.083	0.9335
	Bottom Water Temperature (3-Day Average)	61	-0.096	0.022	-0.429	0.668
	Sediment Temperature (3-Day Average)	61	-0.013	0.026	-0.511	0.60
	Dissolved Oxygen	61	-0.002	0.005	-0.485	0.628
	Hourly Wind Speed	61	0.007	0.028	0.248	0.804
	Biomass	61	$-6x10^{-4}$	0.001	-0.434	0.664
	Abovewater Volume	61	$-4x10^{-4}$	0.001	-0.801	0.423
	Greenness Chromatic Coordinate	61	-0.148	2.873	-0.051	0.959
	Sample Date	61	0.004	0.003	1.220	0.223
Ebullitive CH_4 Flux (Open Sites)	**Depth**	**43**	**-0.014**	**0.007**	**-2.018**	**0.043**
	Bottom Water Temperature (3-Day Average)	37	0.078	0.060	1.285	0.199
	Sediment Temperature (3-Day Average)	37	0.1073	0.076	1.421	0.155
	Dissolved Oxygen	37	-0.006	0.014	-0.426	0.670

	Hourly Wind Speed	37	-0.084	0.082	-1.027	0.304
	Sample Date	95	0.001	0.005	0.195	0.846
Ebullitive CH_4 Flux (Vegetated Sites)	Depth	43	-0.003	0.012	-0.264	0.791
	Bottom Water Temperature (3-Day Average)	36	-0.096	0.108	-0.894	0.371
	Sediment Temperature (3-Day Average)	36	-0.113	0.126	-0.899	0.368
	Dissolved Oxygen	36	-0.012	0.019	-0.628	0.530
	Hourly Wind Speed	36	0.089	0.108	0.831	0.406
	Sample Date	95	-0.004	0.008	-0.533	0.594

Table S3 – Potential CH_4 Production GLMM and post-hoc emmeans results. All tests were run with the cleaned potential CH_4 production dataset which removed the lowest and highest producing vials on each sampling day. Replicates were used as a random effect to account for repeated sampling.

Potential CH_4 Production ~ Treatment + Sample Day + Temperature Control + (1|replicate)					
Parameter	***DF***	***Sum of Squares***	***Mean Squares***	***F Value***	***p-value***
Treatment	3	16.96	5.65	46.87	$< 2e^{-16}$
Sample Day	4	74.89	18.72	155.28	$< 2e^{-16}$
Temperature Control	1	4.67	4.67	38.75	$2.09e^{-4}$
Interactions					
Treatment : Sample Day	12	52.81	4.401	36.49	$< 2e^{-16}$
Treatment : Temperature Control	3	6.97	2.322	19.25	$1.71e^{-9}$
Sample Day : Temperature Control	4	13.89	3.472	28.79	$7.91e^{-15}$
Treatment : Sample Day : Temperature Control	12	24.06	2.005	16.63	$< 2e^{-16}$
Residuals	80	9.65	0.121		

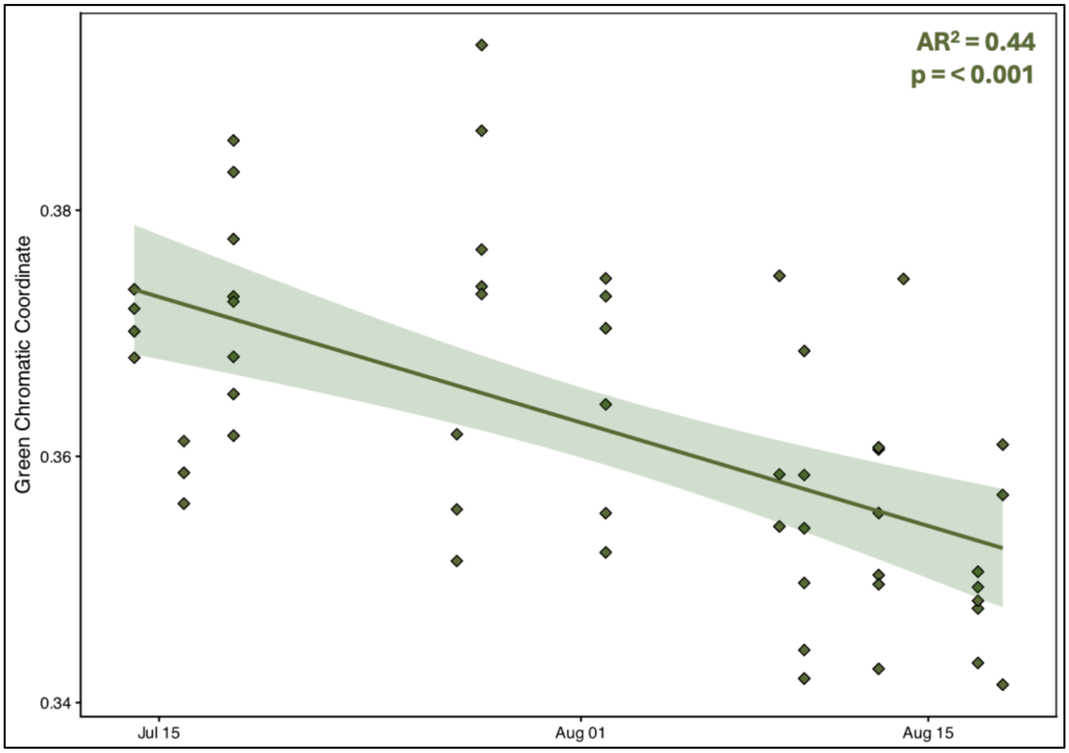

Figure S5 - Greenness Chromatic Coordinate Time Series across field sampling campaign (July 14-August 18). General trend in declining greenness from the beginning to the end of the campaign, with time explaining 32% of the variability in GCC change.

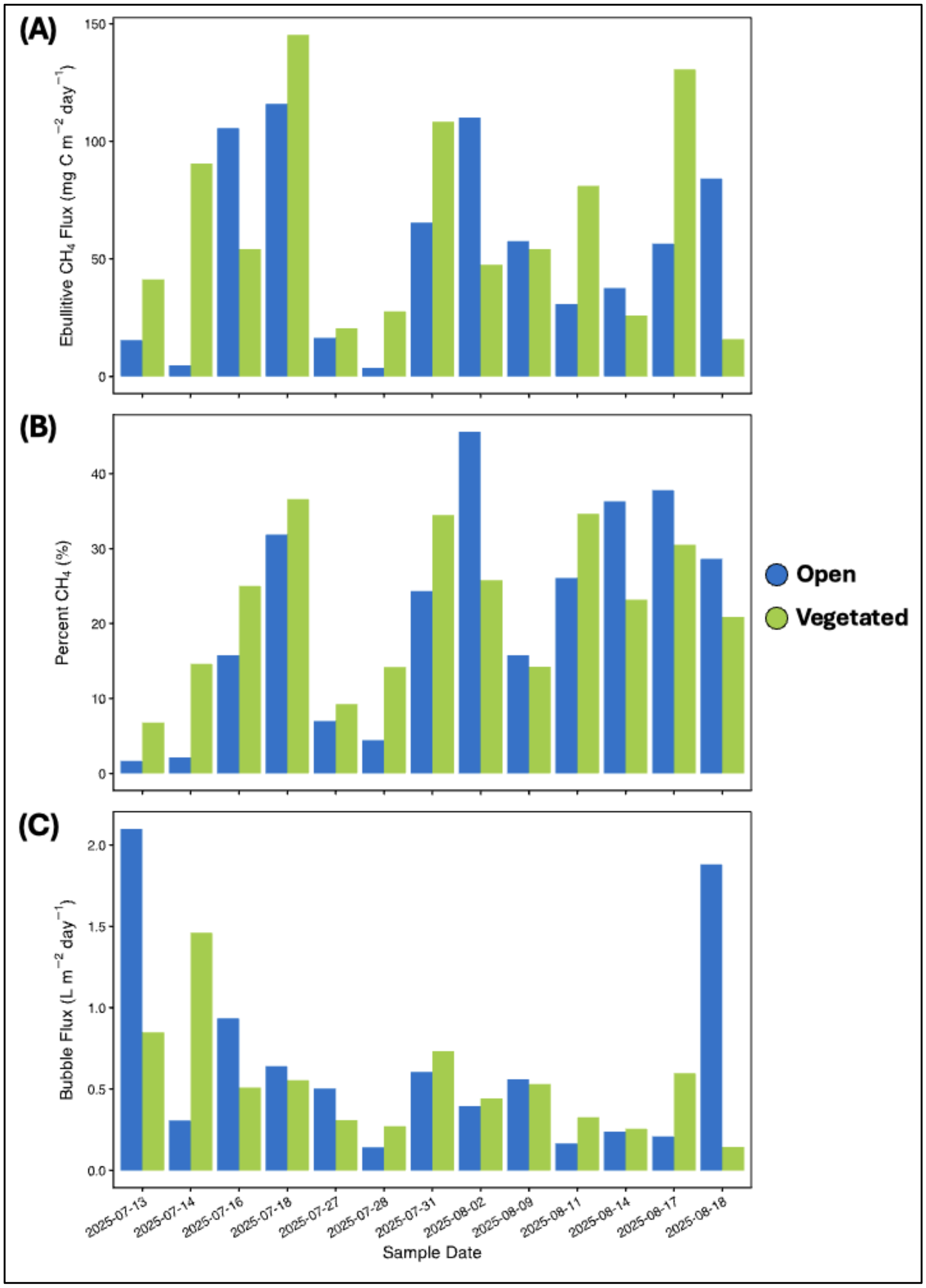

Figure S6 – Ebullitive Flux (A), Percent CH_4 (B), and Bubble Flux (C) across the field season split by open water and vegetated sites.

Table S4 – Progression of CH_4 production concentrations across the 10-day incubation. Incubation results are split into treatment concentrations within each temperature control. Values within the cell are the averaged production rates per sample day in nmol CH_4 g DW^{-1} d^{-1} with their standard deviation provided in brackets (n = 40).

	20°C Treatment Comparisons			30°C Treatment Comparisons		
Sample Day	***S***	***SB***	***B***	***S***	***SB***	***B***
1	0.299 (0.004)	0.408 (0.140)	0.132 (0.004)	0.561 (0.158)	1.54 (0.196)	1.01 (0.789)
2	0.172 (0.112)	1.08 (1.13)	1.05 (0.719)	0.601 (0.107)	1.19 (0.527)	1.76 (0.942)
3	0.461 (0.196)	0.214 (0.235)	3.60 (4.75)	0.629 (0.224)	4.13 (0.01)	3.36 (1.39)
7	2.41 (1.73)	2.38 (1.81)	2.59 (2.48)	2.59 (1.69)	8.79 (3.41)	9.87 (3.33)
10	34.9 (9.2)	30.0 (9.8)	46.6 (12.8)	30.7 (20.3)	71.2 (13.5)	178 (42.1)

Table S5 - Significant differences in potential CH_4 production rates between treatments from the 30°C control based on daily pairwise comparisons (emmeans). Within each cell is the p-value associated with the difference based on emmeans pairwise post-hoc testing, with NS denoting a non-significant result. Statistical significance was assessed at a 95% confidence level (α = 0.05).

	20°C Treatment Comparisons			30°C Treatment Comparisons		
Sample Day	***B-S***	***B-SB***	***S-SB***	***B-S***	***B-SB***	***S-SB***
1	NS	NS	NS	NS	NS	**0.004**
2	**0.004**	NS	**0.003**	**0.03**	NS	NS
3	**0.001**	**<0.0001**	NS	**<0.0001**	NS	**<0.0001**
7	NS	NS	NS	**0.0001**	NS	**0.0005**
10	NS	NS	NS	**<0.0001**	**0.0095**	**0.019**